

(A never-ending) collection of notes on positive scalar curvature*

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This is a growing collection of reading notes on positive scalar curvature. I may or may not have the ability to keep reading.^a

^aI used ChatGPT for transcribing my notes; the content of the notes is written entirely by me. I'm not sure if my understanding of the articles is correct when I'm writing, so please use these notes at your own risk.

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1 Spin and scalar curvature in the presence of a fundamental group. I

1.1 Torus positive scalar curvature

Manifolds are smooth and without boundary.

Theorem 1.1 ([GL80, Theorem A]). *Let X be an enlargeable manifold in the sense of Gromov–Lawson. X carries no Riemannian metric of positive scalar curvature.*

Corollary 1.2 ([GL80, Corollary A]). *Any metric of nonnegative scalar curvature on T^n is flat.*

$X = X_0 \times S^2$, the product of any compact manifold X_0 with S^2 , carries a metric of positive scalar curvature no matter how large $\pi_1(X_0 \times S^2) = \pi_1(X_0)$ is [GL80].

Let X and Y be Riemannian manifolds with metrics g_X and g_Y . For $\varepsilon > 0$, the metric $g = \varepsilon g_X + g_Y$ on $X \times Y$ has scalar curvature $\kappa = \varepsilon^{-1} \kappa_X + \kappa_Y$.

Take $X = S^2(1)$, $Y = X_0$, and $g = \varepsilon g_{S^2(1)} + g_{X_0}$. Since $\kappa_{S^n(1)} = n(n-1)$, setting $n = 2$ gives $\kappa_{S^2(1)} = 2$, and therefore $\kappa = \frac{2}{\varepsilon} + \kappa_{X_0}$. If X_0 is compact, then $\kappa_{X_0} \geq -C \Rightarrow \kappa \geq \frac{2}{\varepsilon} - C$. Hence $\kappa > 0$ for sufficiently small ε .

If $\varepsilon = r^2$, $\kappa_{S^2(r)} = \frac{2}{r^2}$. So $\kappa_{X_0 \times S^2(r)} = \kappa_{X_0} + \frac{2}{r^2}$.

Let $p : \tilde{X} \rightarrow X$ be a covering map, and let g be a metric on X . Set $\tilde{g} = p^*g$. Then $\tilde{g}_{\tilde{x}}(V, W) = g_{p(\tilde{x})}(p_*V, p_*W)$. Hence $p : (\tilde{X}, \tilde{g}) \rightarrow (X, g)$ is a local isometry, so $\kappa_{\tilde{g}}(\tilde{x}) = \kappa_g(p(\tilde{x}))$. Thus $\kappa_{\tilde{g}} \geq \kappa_0 \Rightarrow \kappa_{\tilde{g}} \geq \kappa_0$.

Definition 1.3 ([LM89, Definition IV.5.1]). Let $f : X \rightarrow Y$ be a smooth map between Riemannian manifolds. f is ε -contracting if $\|f_*V\| \leq \varepsilon\|V\|$ for every V tangent to X , where $f_* : T_xX \rightarrow T_{f(x)}Y$.

For every piecewise smooth curve γ in X , $\text{length}(f(\gamma)) \leq \varepsilon \text{length}(\gamma)$ [LM89].

Note 1.4 ([LM89, (IV.5.1)]). For $f : X \rightarrow S^n$,

$$\deg(f) = \frac{\int_X f^* \omega}{\int_{S^n} \omega},$$

where ω is an n -form on S^n with nonzero integral. Here X and S^n are compact oriented manifolds of the same dimension.

The existence of an ε -contracting map $f : X \rightarrow Y$ of nonzero degree means that, in a strong sense, X is bigger than Y by a factor of at least $\frac{1}{\varepsilon}$ [LM89].

A map from a noncompact manifold is constant at infinity if it is constant outside a compact subset. For a map $f : X \rightarrow S^n$ from an oriented n -manifold that is constant at infinity, the formula in Note 1.4 defines $\deg(f)$ [LM89].

Definition 1.5 ([LM89, Definition IV.5.2]). A compact Riemannian n -fold is enlargeable if, for every $\varepsilon > 0$, there is an orientable Riemannian covering $\tilde{X} \rightarrow X$ that admits an ε -contracting map $f : \tilde{X} \rightarrow S^n(1)$ that is onto, constant at infinity, and of nonzero degree.

If the covering can be chosen finite for every $\varepsilon > 0$, then X is compactly enlargeable [LM89].

$\mathbb{R}^n/(k\mathbb{Z})^n \rightarrow \mathbb{R}^n/\mathbb{Z}^n$ is a k^n -fold covering, with $\mathbb{R}^n/(k\mathbb{Z})^n = \tilde{T}^n$ and $\mathbb{R}^n/\mathbb{Z}^n = T^n$. With the standard flat metrics, the map $[x] \mapsto [x/k]$ is an orientation-preserving diffeomorphism $\tilde{T}^n \rightarrow T^n$ and is $\frac{1}{k}$ -contracting. Composing it with a fixed smooth map $T^n \rightarrow S^n(1)$ of degree 1, whose differential is bounded by compactness, gives a map $f : \tilde{T}^n \rightarrow S^n(1)$ of degree 1 that is ε -contracting for sufficiently large k . So T^n is compactly enlargeable, and this property is independent of the Riemannian metric [LM89, proof of Theorem IV.5.3].

1.2 Clifford algebra

Let V be a vector space over a commutative field k , and let q be a quadratic form on V . Its tensor algebra is $\mathcal{T}(V) = \bigoplus_{r=0}^{\infty} \bigotimes^r V$. The ideal $I_q(V)$ in $\mathcal{T}(V)$ is generated by $v \otimes v + q(v)1$, $v \in V$ [LM89, Chapter I, §1].

$\text{Cl}(V, q) = \mathcal{T}(V)/I_q(V)$. Its defining relation is $v \cdot v = -q(v)1$. Assume $\text{char}(k) \neq 2$. Then $v \cdot w + w \cdot v = -2q(v, w)$, where $2q(v, w) = q(v + w) - q(v) - q(w)$. In the positive-definite case, if $e_1, \dots, e_n$ is a q -orthonormal basis, then $e_j^2 = -1$ and $e_j e_k = -e_k e_j$ for $j \neq k$ [LM89, (I.1.3)–(I.1.4)].

$$\text{Cl}^\times(V, q) = \{\varphi \in \text{Cl}(V, q) : \exists \varphi^{-1}, \varphi^{-1}\varphi = \varphi\varphi^{-1} = 1\} \text{ [LM89, (I.2.1)].}$$

Definition 1.7 ([LM89, (I.1.8); Definition I.2.3]). The Pin group $\text{Pin}(V, q)$ is the subgroup of $\text{Cl}^\times(V, q)$ generated by $v \in V$ with $q(v) = \pm 1$.

$\text{Spin}(V, q) = \text{Pin}(V, q) \cap \text{Cl}^0(V, q)$. Here $\text{Cl}^0(V, q)$ denotes the even part of $\text{Cl}(V, q)$.

For the standard positive-definite form on $\mathbb{R}^n$, write $\text{Spin}_n = \text{Spin}(\mathbb{R}^n, \|\cdot\|^2)$ [LM89, (I.2.30)–(I.2.32)].

Let $\pi : E \longrightarrow X$ be a real oriented n -dimensional vector bundle over X , equipped with a Riemannian structure. $P_{\text{SO}_n}(E)$ is the bundle of oriented orthonormal frames. Assume $n \geq 3$.

Definition 1.8 ([LM89, Definition II.1.3 and the diagram of fibrations]). A spin structure on E is a principal Spin_n -bundle $P_{\mathrm{Spin}_n}(E)$ together with a 2-sheeted covering $\xi : P_{\mathrm{Spin}_n}(E) \longrightarrow P_{\mathrm{SO}_n}(E)$ such that $\xi(pg) = \xi(p)\xi_0(g)$ for all $p \in P_{\mathrm{Spin}_n}(E)$ and $g \in \mathrm{Spin}_n$.

$$\begin{array}{ccccc}
& & \text{Spin}_n & \xrightarrow{\xi_0} & \text{SO}_n \\
\mathbb{Z}_2 \swarrow & & \downarrow & & \downarrow \\
& & P\text{Spin}_n(E) & \xrightarrow{\xi} & P\text{SO}_n(E) \\
& & \searrow \pi' & & \swarrow \pi \\
& & X & &
\end{array}$$

ξ_0 is the connected 2-fold covering, and ξ is a 2-sheeted covering.

For $n = 2$, the definition is analogous, using the connected 2-fold covering $\mathrm{SO}_2 \longrightarrow \mathrm{SO}_2$; for $n = 1$, a spin structure is a 2-fold covering of $X = P_{\mathrm{SO}_1}(E)$.

Theorem 1.9 ([LM89, Theorem II.1.7]). *There exists a spin structure on E if and only if the second Stiefel–Whitney class of E is zero. If $w_2(E) = 0$, the distinct spin structures on E are in one-to-one correspondence with the elements of $H^1(X; \mathbb{Z}_2)$.*

A spin manifold is an oriented Riemannian manifold with a spin structure on its tangent bundle [LM89, Chapter II, §2]. For the torus, $TT^n \cong T^n \times \mathbb{R}^n$ is trivial, so all its positive-degree Stiefel–Whitney classes vanish; in particular, $w_2(TT^n) = 0$. By Theorem 1.9, T^n is spin. Similarly, $\mathbb{R}^n/(k\mathbb{Z})^n$ is spin. Thus T^n is enlargeable.

1.3 Complex spinor bundle

Let X be a compact Riemannian spin manifold of dimension $2n$. The Clifford algebra bundle $\text{Cl}(X)$ has fibre $\text{Cl}(T_x X)$ at x . $\mathcal{S}_{\mathbb{C}}(X) = P_{\text{Spin}_{2n}}(X) \times_{\mu_{\mathbb{C}}} M_{\mathbb{C}}$, where $M_{\mathbb{C}} \cong \mathbb{C}^{2^n}$ is the irreducible complex module over $\text{Cl}(\mathbb{R}^{2n})$ and $\mu_{\mathbb{C}} : \text{Spin}_{2n} \rightarrow \text{GL}_{\mathbb{C}}(M_{\mathbb{C}})$ is the restriction of its Clifford action to Spin_{2n} [GL80].

The Riemannian connection on $P_{\text{SO}_{2n}}(X)$ lifts through $P_{\text{Spin}_{2n}}(X) \rightarrow P_{\text{SO}_{2n}}(X)$ and induces a connection on $\mathcal{S}_{\mathbb{C}}(X)$. $\mathcal{S}_{\mathbb{C}}(X)$ is a complex Dirac bundle [LM89, discussion following Theorem II.4.17; Example II.5.9]. This means that unit tangent vectors act unitarily and the metric connection satisfies $\nabla_V(\varphi \cdot \sigma) = (\nabla_V \varphi) \cdot \sigma + \varphi \cdot \nabla_V \sigma$ for sections φ of $\text{Cl}(X)$ and σ of $\mathcal{S}_{\mathbb{C}}(X)$ [LM89, Definition II.5.2].

1.3.1 Some results

- $D\sigma = \sum_{j=1}^{2n} e_j \cdot \nabla_{e_j} \sigma$, where $e_1, \dots, e_{2n}$ are local orthonormal vector fields. This is the Atiyah–Singer operator [LM89, (II.5.0); Example II.5.9].
- For compactly supported sections σ, τ , $\langle D\sigma, \tau \rangle_{L^2} = \langle \sigma, D\tau \rangle_{L^2}$ [LM89, Proposition II.5.3].
- D is elliptic [LM89, Lemma II.5.1].

Let S be a Dirac bundle over X with connection ∇ . $\nabla_{V,W}^2 \sigma = \nabla_V \nabla_W \sigma - \nabla_{\nabla_V W} \sigma$. $\nabla_{V,W}^2 - \nabla_{W,V}^2 = R_{V,W}$ [GL80].

$$\nabla^* \nabla \sigma = -\text{trace}(\nabla^2 \sigma) = -\sum_{j=1}^{2n} \nabla_{e_j, e_j}^2 \sigma.$$

$\langle \nabla^* \nabla \sigma, \tau \rangle_{L^2} = \langle \nabla \sigma, \nabla \tau \rangle_{L^2}$. $\langle \nabla^* \nabla \sigma, \sigma \rangle_{L^2} = \|\nabla \sigma\|_{L^2}^2 \geq 0$; if this is zero, then $\nabla \sigma = 0$. So $\ker(\nabla^* \nabla) = \{\text{parallel sections}\}$ [GL80].

1.3.2 Bochner formula

Define the curvature endomorphism $\mathcal{R} : S \rightarrow S$ by $\mathcal{R}(\sigma) = \frac{1}{2} \sum_{j,k=1}^{2n} e_j \cdot e_k \cdot R_{e_j, e_k}(\sigma)$, where $R_{V,W} : S_x \rightarrow S_x$.

Theorem 1.10 ([GL80, Theorem 1.1 and (1.1)]).

$$D^2 = \nabla^* \nabla + \mathcal{R}. \quad (1.1)$$

Apply Theorem 1.10 to $\mathcal{S}_{\mathbb{C}}(X)$.

$$R_{V,W}^S(\sigma) = \frac{1}{2} \sum_{a < b} \langle R_{V,W}(e_a), e_b \rangle e_a \cdot e_b \cdot \sigma,$$

where $R_{V,W}$ is the Riemann curvature tensor of X , and the products denote Clifford multiplication. Then $\mathcal{R} = \frac{1}{4}\kappa$, and (1.1) becomes

$$D^2 = \nabla^* \nabla + \frac{1}{4}\kappa. \quad (1.2)$$

This is the Lichnerowicz formula. Thus $\kappa > 0 \Rightarrow \ker(D) = 0$ [GL80].

Now let $E \rightarrow X$ be a Hermitian vector bundle equipped with a unitary connection ∇^E . $\mathcal{S}_{\mathbb{C}}(X) \otimes_{\mathbb{C}} E$ is a bundle of modules over $\text{Cl}(X)$ and is also a complex Dirac bundle. $\nabla_V(\sigma \otimes e) = (\nabla_V^S \sigma) \otimes e + \sigma \otimes \nabla_V^E e$ [LM89, Proposition II.5.10; note preceding Theorem II.8.17].

D_E is the Dirac operator of the twisted spinor bundle. $D_E(\sigma \otimes e) = \sum_{j=1}^{2n} e_j \cdot \nabla_{e_j}(\sigma \otimes e)$.

$$R_{e_j, e_k}^{S \otimes E} = R_{e_j, e_k}^S \otimes 1 + 1 \otimes R_{e_j, e_k}^E \cdot \frac{1}{2} \sum_{j,k} e_j \cdot e_k \cdot (R_{e_j, e_k}^S \otimes 1) = \frac{1}{4}\kappa.$$

Following [GL80, (1.3)], define $\mathcal{R}^E : \mathcal{S}_{\mathbb{C}}(X) \otimes_{\mathbb{C}} E \rightarrow \mathcal{S}_{\mathbb{C}}(X) \otimes_{\mathbb{C}} E$ by

$$\mathcal{R}^E(\sigma \otimes e) = \frac{1}{2} \sum_{j,k=1}^{2n} (e_j \cdot e_k \cdot \sigma) \otimes R_{e_j, e_k}^E(e). \quad (1.3)$$

Theorem 1.11 ([LM89, Theorem II.8.17 and (II.8.23)]).

$$D_E^2 = \nabla^* \nabla + \frac{1}{4}\kappa + \mathcal{R}^E. \quad (1.4)$$

Below, endomorphism norms are the suprema of the fibrewise operator norms; curvature two-form norms also take the supremum over unit tangent vectors in both slots.

Theorem 1.12 ([GL80, Theorem 1.3]; compare [LM89, (IV.5.5)]). *Let X be a compact Riemannian spin manifold, and let $\mathcal{S}_{\mathbb{C}}(X) \otimes_{\mathbb{C}} E$ be a twisted spinor bundle. If $\kappa \geq \kappa_0 > 0$ and $\|\mathcal{R}^E\| < \frac{\kappa_0}{4}$, then $\ker D_E = 0$.*

From $\mathcal{S}_{\mathbb{C}}(X) \otimes_{\mathbb{C}} E$, one can construct a natural elliptic complex.

Let $e_1, \dots, e_{2n}$ be local positively oriented orthonormal vector fields. $\omega_{\mathbb{C}} = i^n e_1 \cdots e_{2n}$ is a parallel section of $\text{Cl}(X) \otimes_{\mathbb{R}} \mathbb{C}$, so $\nabla \omega_{\mathbb{C}} = 0$. By the Clifford relations, $\omega_{\mathbb{C}}^2 = 1$ and $e_j \omega_{\mathbb{C}} = -\omega_{\mathbb{C}} e_j$ for every j .

$$\mathcal{S}_{\mathbb{C}}(X) = \underbrace{\mathcal{S}_{\mathbb{C}}^+(X)}_{\{\sigma : \omega_{\mathbb{C}} \cdot \sigma = \sigma\}} \oplus \underbrace{\mathcal{S}_{\mathbb{C}}^-(X)}_{\{\sigma : \omega_{\mathbb{C}} \cdot \sigma = -\sigma\}}. \quad (1.5)$$

Tensoring (1.5) with E gives $\mathcal{S}_{\mathbb{C}}(X) \otimes_{\mathbb{C}} E = (\mathcal{S}_{\mathbb{C}}^+(X) \otimes_{\mathbb{C}} E) \oplus (\mathcal{S}_{\mathbb{C}}^-(X) \otimes_{\mathbb{C}} E)$. Since $\omega_{\mathbb{C}}$ is parallel and anticommutes with tangent vectors, D_E interchanges the two summands. With respect to this splitting,

$$D_E = \begin{pmatrix} 0 & D_E^- \\ D_E^+ & 0 \end{pmatrix}. \quad (1.6)$$

Here $D_E^- : \Gamma(\mathcal{S}_{\mathbb{C}}^-(X) \otimes_{\mathbb{C}} E) \rightarrow \Gamma(\mathcal{S}_{\mathbb{C}}^+(X) \otimes_{\mathbb{C}} E)$.

The twisted Atiyah–Singer operator is $D_E^+ : \Gamma(\mathcal{S}_{\mathbb{C}}^+(X) \otimes_{\mathbb{C}} E) \rightarrow \Gamma(\mathcal{S}_{\mathbb{C}}^-(X) \otimes_{\mathbb{C}} E)$. D_E is formally self-adjoint. $D_E^- = (D_E^+)^*$ [GL80, (1.4)].

1.3.3 Analytic index

If X is compact and D_E^+ is elliptic, then its kernel and cokernel are finite-dimensional. Its analytic index is

$$\text{ind}(D_E^+) = \dim_{\mathbb{C}} \ker D_E^+ - \dim_{\mathbb{C}} \text{coker } D_E^+. \quad (1.7)$$

Ellipticity on a compact manifold implies that the range is closed. So $\text{coker } D_E^+ \cong \ker((D_E^+)^*) \cong \ker(D_E^-)$. By (1.6), $\ker(D_E) = \ker D_E^+ \oplus \ker D_E^-$ [LM89, Theorem III.5.2(ii) and (III.5.2)].

By Theorem 1.12, $\ker D_E = 0 \Rightarrow \ker D_E^+ = 0$, $\ker D_E^- = 0$, and $\text{coker } D_E^+ = 0$. So (1.7) gives $\text{ind}(D_E^+) = 0$. When $\kappa > 0$, take E to be the trivial flat complex line bundle. Then $R^E = 0$, and the same argument using (1.2) gives $\text{ind}(D^+) = 0$ [GL80, Theorem 1.4].

1.3.4 Topological index

Let E be a complex vector bundle of rank d . Its total Chern class is $c(E) = 1 + c_1(E) + \cdots + c_d(E)$, where, formally, $c(E) = \prod_{k=1}^d (1 + x_k)$.

The Chern character is

$$\text{ch}(E) = e^{x_1} + \cdots + e^{x_d} = d + \sum_{j=1}^d x_j + \frac{1}{2} \sum_{j=1}^d x_j^2 + \cdots \in H^{2*}(X; \mathbb{Q}) \quad [\text{LM89, (III.11.22)}].$$

$\text{ch}^k(E) = \frac{1}{k!} \sum_{j=1}^d x_j^k$. $\text{ch}(E \oplus E') = \text{ch}(E) + \text{ch}(E')$. $\text{ch}(E \otimes E') = \text{ch}(E) \text{ch}(E')$ [LM89, (III.11.23) and Proposition III.11.16].

$H^{2j}(S^{2n}; \mathbb{Z}) = 0$ for $0 < j < n$. Let $E_0 \rightarrow S^{2n}(1)$ have rank d . Then $c_j(E_0) = 0$ for $0 < j < n$. If $c_n(E_0) \neq 0$, the formulas [LM89, (III.11.22)–(III.11.23)] give

$$\text{ch}(E_0) = d + \frac{(-1)^{n-1}}{(n-1)!} c_n(E_0). \quad (1.8)$$

Example 1.13 ([LM89, Example III.11.13 and (III.11.17)]). The $\widehat{A}$ -sequence is determined by $\widehat{a}(x) = \frac{\sqrt{x}/2}{\sinh(\sqrt{x}/2)}$. For a real oriented $2m$ -dimensional vector bundle E , write formally $E \otimes_{\mathbb{R}} \mathbb{C} = \ell_1 \oplus \bar{\ell}_1 \oplus \cdots \oplus \ell_m \oplus \bar{\ell}_m$ according to the Splitting Principle and set $x_j = c_1(\ell_j)$. Then

$$\widehat{A}(E) = \prod_{j=1}^m \frac{x_j/2}{\sinh(x_j/2)}, \quad \widehat{A}(E) = 1 + \widehat{A}_1(p_1 E) + \widehat{A}_2(p_1 E, p_2 E) + \cdots.$$

$\widehat{A}(TX) \in H^{4*}(X; \mathbb{Q})$. $\widehat{A}(X) = \{\widehat{A}(TX)\}[X]$, where the top-degree part is evaluated on the fundamental class $[X]$. So $\widehat{A}(X) = 0$ whenever $\dim X \not\equiv 0 \pmod{4}$ [LM89, (III.11.19)].

Recall that $\text{ind}(D_E^+) = \dim_{\mathbb{C}} \ker D_E^+ - \dim_{\mathbb{C}} \text{coker } D_E^+$.

Theorem 1.14 ([LM89, Theorem III.13.10 and (III.13.26)]). *For the twisted Atiyah–Singer operator,*

$$\text{ind}(D_E^+) = \{\text{ch}(E) \widehat{A}(TX)\}[X]. \quad (1.9)$$

For the trivial complex line bundle E , $\text{ch}(E) = 1$, so Theorem 1.14 gives $\underline{\text{ind}}(D^+) = \widehat{A}(X)$.

1.4 Compactly enlargeable spin manifolds

If X is compactly enlargeable, then $X \times S^1$ is compactly enlargeable. If X is spin, then $X \times S^1$ is spin. If X carries a metric with $\kappa > 0$, then $X \times S^1$ carries a metric with $\kappa > 0$ [LM89, Proposition II.2.15; (IV.5.2)–(IV.5.3)].

Let X be a compactly enlargeable spin manifold, and suppose that X carries a Riemannian metric with $\kappa > 0$. Since X is compact, $\kappa \geq \kappa_0 > 0$ for some κ_0 . Replacing X by $X \times S^1$ if necessary, assume that X has dimension $2n$.

Choose a complex vector bundle $E_0 \rightarrow S^{2n}(1)$ of rank d such that $c_n(E_0) \neq 0$. Such a bundle exists by the Chern-character isomorphism for the even-dimensional sphere [LM89, proof of Theorem IV.5.5]. Fix a Hermitian metric and a unitary connection ∇^{E_0} on E_0 . Let R^{E_0} be its curvature 2-form. Its norm is finite because the sphere is compact.

Let $\varepsilon > 0$. Since X is compactly enlargeable, there is a finite orientable Riemannian covering $p : \widetilde{X} \rightarrow X$ that admits an ε -contracting map $f : \widetilde{X} \rightarrow S^{2n}(1)$ of nonzero degree. The metric on $\widetilde{X}$ is the lifted metric p^*g . Since p is a local isometry, $\kappa_{\widetilde{X}} = \kappa_X \circ p$, so $\kappa_{\widetilde{X}} \geq \kappa_0$ [LM89, Proof of Theorem IV.5.5].

Because X is spin, the pullback spin structure makes $\widetilde{X}$ spin; equivalently, $w_2(T\widetilde{X}) = p^*w_2(TX) = 0$. Thus the twisted Atiyah–Singer operator can be constructed on $\widetilde{X}$.

Let $E = f^*E_0$, $\nabla^E = f^*\nabla^{E_0}$. Under the identification $E_x = (E_0)_{f(x)}$, $R_{V,W}^E = R_{f_*V, f_*W}^{E_0}$.

Consider $\mathcal{S}_{\mathbb{C}}(\widetilde{X}) \otimes_{\mathbb{C}} E$. By Theorem 1.11, its Bochner formula is (1.4), where $\mathcal{R}^E$ is given by (1.3) and depends universally and linearly on the components of R^E . Thus, by [LM89, (IV.5.4) and (IV.5.7)], there is a constant k_n depending only on the dimension such that

$$\|\mathcal{R}^E\| \leq k_n \|R^E\|. \quad (1.10)$$

Moreover, $\|\mathcal{R}^E\| \leq \varepsilon^2 \|R^{E_0}\|$. As in [LM89, (IV.5.8)], combining (1.10) with this estimate gives

$$\|\mathcal{R}^E\| \leq k_n \varepsilon^2 \|R^{E_0}\| < \frac{\kappa_0}{4} \quad \text{for sufficiently small } \varepsilon;$$

$$\text{equivalently, } 0 < \varepsilon < \sqrt{\frac{\kappa_0}{4k_n \|R^{E_0}\|}}.$$

Then $\|\mathcal{R}^E\| < \frac{\kappa_0}{4}$. Since $\kappa \geq \kappa_0$, for every nonzero σ in a fibre of the twisted spinor bundle,

$$\left\langle \left(\frac{1}{4} \kappa + \mathcal{R}^E \right) \sigma, \sigma \right\rangle \geq \left(\frac{\kappa_0}{4} - \|\mathcal{R}^E\| \right) |\sigma|^2 > 0.$$

Integrating and using (1.4) and $\nabla^* \nabla \geq 0$, we get $D_E^2 > 0$, so $\ker D_E = 0$. Hence $\ker D_E^+ = 0$ and $\ker D_E^- = 0$. Since $\text{coker } D_E^+ \cong \ker D_E^-$, $\text{ind}(D_E^+) = 0$ on the analytic side.

By the Atiyah–Singer formula (1.9) and (1.8), using $\text{ch}(E) = f^* \text{ch}(E_0) = d + \frac{(-1)^{n-1}}{(n-1)!} c_n(E)$,

$$\begin{aligned}
 \text{ind}(D_E^+) &= \{\text{ch}(E) \widehat{A}(T\widetilde{X})\} [\widetilde{X}] \\
 &= \left\{ \left(d + \frac{(-1)^{n-1}}{(n-1)!} c_n(E) \right) \widehat{A}(T\widetilde{X}) \right\} [\widetilde{X}] \\
 &= d \underbrace{\widehat{A}(\widetilde{X})}_{\text{ind}(D^+) = 0} + \frac{(-1)^{n-1}}{(n-1)!} c_n(E) [\widetilde{X}] \\
 &= \frac{(-1)^{n-1}}{(n-1)!} c_n(E) [\widetilde{X}] \\
 &= \frac{(-1)^{n-1}}{(n-1)!} \underbrace{f^* c_n(E_0) [\widetilde{X}]}_{\substack{= \deg(f) c_n(E_0) [S^{2n}] \\ \deg(f) \neq 0}} \neq 0.
 \end{aligned}$$

See [LM89, (IV.5.9); (III.11.22)–(III.11.23)].

So a compactly enlargeable spin manifold cannot carry a metric with positive scalar curvature [GL80, Proposition 3.1; Theorem 3.7]; compare [LM89, Theorem IV.5.5].

1.5 The torus

By Example 1.6, T^n is compactly enlargeable; T^n and its finite covering tori are spin.

- (1) $n = 2m$. Suppose $\kappa > 0$, so $\kappa \geq \kappa_0 > 0$ for some κ_0 . Compact enlargeability gives a finite covering $p : \widetilde{T}^{2m} \rightarrow T^{2m}$ and an ε -contracting map $f : \widetilde{T}^{2m} \rightarrow S^{2m}(1)$ of nonzero degree for $\varepsilon > 0$ sufficiently small. Since $\kappa_{\widetilde{T}^{2m}} \geq \kappa_0$ and $\widetilde{T}^{2m}$ is spin, the preceding contradiction applies. Thus T^{2m} carries no metric with $\kappa > 0$.
- (2) $n = 2m + 1$. Suppose T^{2m+1} carries a metric g with $\kappa_g > 0$. $T^{2m+1} \times S^1 = T^{2m+2}$.

$$\kappa_{g+dt^2} = \kappa_g + \kappa_{S^1} = \kappa_g > 0,$$

which contradicts the even-dimensional case above [LM89, (IV.5.2)–(IV.5.3)].

Theorem 1.15 ([LM89, Theorem IV.5.7 (J. P. Bourguignon)]). *Let X be a compact connected manifold that carries no metric with $\kappa > 0$. Then every metric on it with $\kappa \geq 0$ is Ricci flat.*

Proposition 1.16 ([LM89, Proposition IV.5.8]). *A Ricci-flat enlargeable manifold is flat.*

Applying Theorem 1.15 and Proposition 1.16 to T^n gives Corollary 1.2.

2 A proof of Gromov's cube inequality on scalar curvature

2.1 Cube inequality

Theorem 2.1 ([WXY23, Theorem 1.1]). *Let g be a Riemannian metric on the cube $I^n = [0, 1]^n$, $n \geq 2$. Suppose that $\ell_i = \text{dist}(\partial_{i-}, \partial_{i+})$ is the g -distance between the i -th pair of opposite faces ∂_{i-} and ∂_{i+} of the cube. Then:*

(1) *If $\text{Sc}_g \geq k > 0$, then*

$$\sum_{i=1}^n \frac{1}{\ell_i^2} \geq \frac{kn}{4\pi^2(n-1)}.$$

Consequently,

$$\min_{1 \leq i \leq n} \text{dist}(\partial_{i-}, \partial_{i+}) \leq 2\pi \sqrt{\frac{n-1}{k}}.$$

(2) *If $\text{Sc}_g \geq k > 0$ and the dihedral angles of (I^n, g) are $< \pi$, then*

$$\sum_{i=1}^n \frac{1}{\ell_i^2} > \frac{kn}{4\pi^2(n-1)}.$$

Consequently,

$$\min_{1 \leq i \leq n} \text{dist}(\partial_{i-}, \partial_{i+}) < 2\pi \sqrt{\frac{n-1}{k}}.$$

Remark 2.2. A point of I^n is $x = (x_1, \dots, x_n)$, where $0 \leq x_i \leq 1$.

$$\partial_{i-} = \{x \in I^n : x_i = 0\} \quad \partial_{i+} = \{x \in I^n : x_i = 1\}.$$

For $I^2 = [0, 1]^2$,

$$\begin{aligned} \partial_{1-} &= \{0\} \times [0, 1], & \partial_{1+} &= \{1\} \times [0, 1], \\ \partial_{2-} &= [0, 1] \times \{0\}, & \partial_{2+} &= [0, 1] \times \{1\}. \end{aligned}$$

Proof. Part I: $\geq$ inequality

By rescaling the metric, it suffices to assume $k = n(n-1)$.

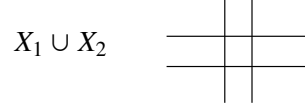
Step 1. Assume to the contrary that $\sum_{i=1}^n \frac{1}{\ell_i^2} < \frac{n^2}{4\pi^2}$. Consequently, $\sum_{i=1}^n \frac{\pi^2}{\ell_i^2} < \frac{n^2}{4}$.

As $\varepsilon \rightarrow 0$, $\frac{\pi^2(1+\varepsilon)^2}{(\ell_i - 2\varepsilon)^2} \rightarrow \frac{\pi^2}{\ell_i^2}$. This depends continuously on ε . Hence there exists $\varepsilon > 0$ such that $0 < 2\varepsilon < \min_i \ell_i$, $\varepsilon < \min_i \pi^2 / \ell_i^2$, and

$$\sum_{i=1}^n \frac{\pi^2 (1+\varepsilon)^2}{(\ell_i - 2\varepsilon)^2} < \frac{n^2}{4}.$$

Step 2. Extend the metric on I^n to a complete metric on $\mathbb{R}^n$.

(a) The case $n = 2$. Let $X = [0, 1]^2$, $X_1 = [0, 1] \times \mathbb{R}$, and $X_2 = \mathbb{R} \times [0, 1]$. Then $X_1 \cap X_2 = X$.



Original metric: $g = \sum_{a,b=1}^2 g_{ab} dx_a dx_b$.

g on $[0, 1]^2$:

$$\ell_1 = \text{dist}_g(\{0\} \times [0, 1], \{1\} \times [0, 1]),$$

$$\ell_2 = \text{dist}_g([0, 1] \times \{0\}, [0, 1] \times \{1\}).$$

Whitney's extension theorem and a smooth partition of unity extend the smooth coefficient functions, preserve a positive-definite symmetric form, and transition to the Euclidean metric. Thus $g_1 = g$ on X , while $g_1 = dx_1^2 + dx_2^2$ outside a compact set of $X_1 \cup X_2$.

$$\begin{aligned} \partial_- X_1 &= \{0\} \times \mathbb{R} & \partial_+ X_1 &= \{1\} \times \mathbb{R} \\ \partial_- X_2 &= \mathbb{R} \times \{0\} & \partial_+ X_2 &= \mathbb{R} \times \{1\}, \end{aligned}$$

Difficulty: g_1 may create short curves outside X .

Choose a smooth positive function $\bar{\Psi}$ such that $\bar{\Psi} = 1$ on X , and $\bar{\Psi}$ is constant when $x_1^2 + x_2^2$ is sufficiently large. $g_2 := \bar{\Psi} g_1$.

$$L_{g_2}(\gamma) = \int \sqrt{\bar{\Psi}(\gamma(t))} |\gamma'(t)|_{g_1} dt.$$

Choose $\bar{\Psi}$ large enough so that, on $X_1 \cup X_2$, g_2 satisfies

$$(1) \quad \text{dist}_{g_2}(\partial_- X_1, \partial_+ X_1) \geq \ell_1 - \frac{\varepsilon}{3},$$

$$(2) \quad \text{dist}_{g_2}(\partial_- X_2, \partial_+ X_2) \geq \ell_2 - \frac{\varepsilon}{3}$$

$$g_2 = \underbrace{R^2}_{\bar{\Psi} \text{ constant}} (dx_1^2 + dx_2^2) \quad \text{outside of a compact subset of } X_1 \cup X_2.$$

Extend g_2 to a **complete** metric $\bar{g}$ on $\mathbb{R}^2$ such that

$$\bar{g} = \underbrace{R^2}_{\text{constant}} \underbrace{(dx_1^2 + dx_2^2)}_{\text{Euclidean}} \quad \text{outside a compact subset of } \mathbb{R}^2.$$

Thus $\bar{g}$ is flat and $\text{Sc}_{\bar{g}} = 0$ outside a compact set. On the remaining compact set, $\text{Sc}_{\bar{g}}$ is continuous, so it has a finite minimum. **Continuity $\Rightarrow$ finite minimum.** Choose $\sigma > 0$ sufficiently large so that $\text{Sc}_{\bar{g}} \geq -\sigma$ on $\mathbb{R}^2$.

(b) *General construction.*

For $k = 0, 1, \dots, n$, Λ_k is the family of all subsets of $\{1, \dots, n\}$ with k elements. For $\lambda \in \Lambda_k$, set $X_\lambda = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n : 0 \leq x_j \leq 1 \text{ if } j \notin \lambda\}$. Thus the coordinates indexed by λ are unrestricted, while the other coordinates lie in $[0, 1]$. $X_\emptyset = [0, 1]^n$.

For $j \notin \lambda$, the faces on $x_j = 0$ and $x_j = 1$ are $\partial_{j-}X_\lambda$ and $\partial_{j+}X_\lambda$, respectively.

$$\partial_{j-}X_\lambda = \{x_j = 0\} \cap X_\lambda, \quad \partial_{j+}X_\lambda = \{x_j = 1\} \cap X_\lambda.$$

We construct the metric by induction so that, for $0 \leq k \leq n-1$, $\lambda \in \Lambda_k$, and $j \notin \lambda$,

$$\text{dist}_{\bar{g}}(\partial_{j-}X_\lambda, \partial_{j+}X_\lambda) \geq \ell_j - \frac{k\varepsilon}{n+1} \quad (2.3)$$

These distances are intrinsic in X_λ ; the case $k = 0$ holds by definition. At each of the k stages, each remaining face distance decreases by at most $\frac{\varepsilon}{n+1}$, giving $\frac{k\varepsilon}{n+1}$ after k stages.

Induction step: k to $k+1$.

Assume the metric has been constructed on the union of X_λ , $\lambda \in \Lambda_k$, satisfying (2.3). Let $\mu \in \Lambda_{k+1}$. The $k+1$ sets X_λ , where $\lambda \subset \mu$ and $|\lambda| = k$, are already contained in X_μ :

$$\bigcup_{\substack{\lambda \subset \mu \\ |\lambda|=k}} X_\lambda \subset X_\mu.$$

When two such sets overlap, their intersection is an X_λ from an earlier stage. Extend the metric from this union to X_μ using Whitney's extension theorem, a smooth partition of unity, and rescaling; an additional coordinate ranges over $\mathbb{R}$. At this stage, the loss changes from $k\varepsilon/(n+1)$ to $(k+1)\varepsilon/(n+1)$. After the final extension, $\bar{g} = g$ on I^n and $\bar{g} = R^2 g_o$ outside a compact set, where g_o is the standard Euclidean metric and $R > 0$. Thus $\bar{g}$ is complete; choose $\sigma > 0$ so that $\text{Sc}_{\bar{g}} \geq -\sigma$.

Let $E_i = \{(y_1, \dots, y_n) \in \mathbb{R}^n : y_i = 0\}$.

$$\text{1-Lipschitz} \longrightarrow \varphi_i = \text{sgn}(x_i) \text{dist}_{\bar{g}}(x, E_i) = \begin{cases} \text{dist}_{\bar{g}}(x, E_i) & x_i > 0 \\ 0 & x_i = 0 \\ -\text{dist}_{\bar{g}}(x, E_i) & x_i < 0 \end{cases}$$

1-Lipschitz: $|\text{dist}_{\bar{g}}(x, E_i) - \text{dist}_{\bar{g}}(y, E_i)| \leq \text{dist}_{\bar{g}}(x, y)$. For points on opposite sides, every connecting path crosses E_i , so the sum of their distances to E_i is at most their mutual distance. Hence the signed function φ_i is also 1-Lipschitz.

$$\underbrace{\psi_i}_{\text{Centered.}} = \varphi_i - \frac{\ell_i}{2}, \quad x_i = 0 \Rightarrow \psi_i(x) = -\ell_i/2.$$

By smooth Lipschitz approximation, choose a smooth z_i such that

$$|z_i - \underbrace{\psi_i}_{1\text{-Lipschitz}}| \leq \frac{\varepsilon}{n+1}, \quad \|\nabla z_i\| \leq 1 + \varepsilon.$$

If $x_j \geq 1$, every path from x to E_j crosses the slab $0 \leq x_j \leq 1$. Apply (2.3) with $k = n - 1$ to that crossing.

$$\text{If } x_j < 0, \quad \varphi_j(x) \leq 0, \quad \psi_j(x) = \varphi_j(x) - \frac{\ell_j}{2} \leq -\frac{\ell_j}{2},$$

$$z_j(x) \leq -\frac{\ell_j}{2} + \frac{\varepsilon}{n+1}.$$

$$\text{If } x_j > 1, \quad \psi_j(x) \geq \frac{\ell_j}{2} - \frac{(n-1)\varepsilon}{n+1} \quad \text{by (2.3),}$$

$$z_j(x) \geq \frac{\ell_j}{2} - \frac{n\varepsilon}{n+1} > \frac{\ell_j}{2} - \varepsilon.$$

$\|\nabla z_i\| \leq 1 + \varepsilon \rightsquigarrow$ for every vector field V , $|V(z_i)| \leq (1 + \varepsilon)\|V\|$. Since $\bar{g}$ is uniformly equivalent to g_o , $\text{dist}_{\bar{g}}(x, E_i) \geq c|x_i|$ for some $c > 0$. Thus $\sum_i z_i(x)^2 \rightarrow \infty$ as $|x| \rightarrow \infty$; in particular, its sublevel sets are compact.

Step 3. Choose a positive $\delta > 0$ and $\{\bar{\ell}_i\}_{1 \leq i \leq n}$ such that $0 < \bar{\ell}_i < \ell_i - 2\varepsilon < \ell_i$,

$$\frac{n^2}{4} - \sum_{i=1}^n \frac{\pi^2(1+\varepsilon)^2}{(\ell_i - 2\varepsilon)^2} > \delta$$

and, for every $j = 1, \dots, n$,

$$-\frac{n\sigma}{4(n-1)} + \frac{\pi^2(1+\varepsilon)^2}{(\ell_j - 2\varepsilon)^2} \tan^2\left(\frac{\pi\bar{\ell}_j}{2(\ell_j - 2\varepsilon)}\right) - \varepsilon(1+\varepsilon) - \sum_{i \neq j} \frac{\pi^2(1+\varepsilon)^2}{(\ell_i - 2\varepsilon)^2} > \delta$$

$$\bar{\ell}_j \rightarrow \ell_j - 2\varepsilon \quad \text{from below.} \quad \text{So} \quad \frac{\pi\bar{\ell}_j}{2(\ell_j - 2\varepsilon)} \rightarrow \frac{\pi}{2} \rightsquigarrow \tan^2\left(\frac{\pi\bar{\ell}_j}{2(\ell_j - 2\varepsilon)}\right) \rightarrow \infty.$$

Set $r_i := \frac{\pi(1+\varepsilon)}{\ell_i - 2\varepsilon}$. Then $\frac{n^2}{4} - \sum_{i=1}^n r_i^2 > \delta$. Fix $\ell > 0$ such that $\ell > \ell_i/2 + 1$ for every i .

Our choice of ε gives $\varepsilon(1+\varepsilon) < r_i^2$. Choose smooth $\{\xi_i\}_{1 \leq i \leq n}$ on $\mathbb{E}^2$, even in both variables, such that

$$0 \leq (1+\varepsilon)\xi_i(x, y) \leq y^2 + r_i^2,$$

$$0 \leq \xi_i(x, y) \leq \varepsilon \quad \text{for } |x| \geq \frac{\ell_i}{2} - \varepsilon,$$

with smooth interpolation between the specified regions:

$$\xi_i(x, y) = \begin{cases} \frac{y^2 + r_i^2}{1 + \varepsilon}, & |x| \leq \frac{\bar{\ell}_i}{2}, \quad \text{near origin.} \\ 0, & \frac{\ell_i}{2} - \varepsilon \leq |x| \leq \ell, \quad \text{near exterior threshold} \\ \varepsilon, & |x| \geq \ell + 1, \quad \text{far away.} \end{cases}$$

Here $\bar{\ell}_i < \ell_i - 2\varepsilon$, so $\bar{\ell}_i/2 < \ell_i/2 - \varepsilon$. Choose the interpolation to vanish already slightly before $|x| = \ell_i/2 - \varepsilon$; this leaves a nonempty interval after $\bar{\ell}_i/2$ on which to interpolate.

$$\left\{ \begin{array}{l} f_i^2 - (1 + \varepsilon)f_i' \geq -r_i^2 \\ f_i \text{ is large near the relevant face level} \\ f_i' \text{ is small outside the cube} \\ f_i \text{ is defined on } \mathbb{R} \\ \sum_{i=1}^n f_i(z_i)^2 \text{ is proper.} \end{array} \right.$$

Let f_i be a solution of

$$\begin{cases} f'(x) = \xi_i(x, f(x)), \\ f(0) = 0, \end{cases} \quad \xi_i \geq 0 \rightsquigarrow f_i'(x) \geq 0 \Rightarrow f_i \text{ is nondecreasing.}$$

Consider $h_i(x) = r_i \tan\left(\frac{r_i x}{1 + \varepsilon}\right)$:

$$\begin{aligned} h_i'(x) &= \frac{r_i^2}{1 + \varepsilon} \sec^2\left(\frac{r_i x}{1 + \varepsilon}\right) \\ &= \frac{r_i^2}{1 + \varepsilon} \left[1 + \tan^2\left(\frac{r_i x}{1 + \varepsilon}\right) \right] \\ &= \frac{r_i^2 + h_i(x)^2}{1 + \varepsilon} \end{aligned}$$

So h_i solves

$$\begin{cases} f'(x) = \frac{f(x)^2 + r_i^2}{1 + \varepsilon} \\ f(0) = 0, \end{cases}$$

By uniqueness,

$$f_i(x) = r_i \tan\left(\frac{r_i x}{1 + \varepsilon}\right), \quad |x| \leq \frac{\bar{\ell}_i}{2}.$$

The tangent has poles at

$$|x| = \frac{\pi(1 + \varepsilon)}{2r_i} = \frac{\ell_i - 2\varepsilon}{2} = \frac{\ell_i}{2} - \varepsilon.$$

On $\ell_i/2 - \varepsilon \leq |x| \leq \ell$, $f'_i(x) = 0$. For $|x|$ large, $f'_i(x) = \varepsilon$, so f_i is linear. Comparison with h_i bounds f_i until the cutoff becomes zero, strictly before the tangent pole. It is then constant up to $|x| = \ell$, and has bounded derivative beyond that, so there is no finite-time blow-up.

$\sum_{i=1}^n f_i(z_i)^2$ is proper. The function f_i exists on $\mathbb{R}$, and for $|x|$ large it is of the form $\varepsilon x \pm c$. The evenness of ξ_i makes f_i odd. Since $f_i(x) - \varepsilon x$ is bounded and $\sum_i z_i^2$ is proper, $\sum_i f_i(z_i)^2$ is proper as well.

Step 4. $\text{Cl}_{n,0}$ is generated by $\{e_i\}_{1 \leq i \leq n}$ subject to $e_i^2 = -1$, $e_i e_j + e_j e_i = 0$, $i \neq j$.

$\text{Cl}_{0,n}$ is generated by $\{\widehat{e}_i\}_{1 \leq i \leq n}$ subject to $\widehat{e}_i^2 = 1$, $\widehat{e}_i \widehat{e}_j + \widehat{e}_j \widehat{e}_i = 0$, $i \neq j$.

Let $E := S(T\mathbb{R}^n \oplus \mathbb{E}^n)$ be equipped with the action of $\text{Cl}_{n,0}(T\mathbb{R}^n) \widehat{\otimes} \text{Cl}_{0,n}(\mathbb{E}^n)$. Here S is the graded spinor module bundle of rank 2^n , and the auxiliary bundle $\mathbb{E}^n$ has its flat connection. The connection on E is the induced tensor connection. Clifford actions from the two factors anticommute. Let D be the twisted Dirac operator on $L^2(\mathbb{R}^n, E)$. In terms of local orthonormal frames $\{w_i\}_{1 \leq i \leq n}$, $D = \sum_{i=1}^n w_i \nabla_{w_i}$.

$$\begin{aligned}
 B &= \overset{\text{Dirac}}{D \widehat{\otimes} 1} + \sum_{i=1}^n \overset{n \text{ functions } f_i(z_i)}{1 \widehat{\otimes} f_i(z_i) \widehat{e}_i} \quad \text{Callias-type operator.} \\
 B^2 &= (D \widehat{\otimes} 1)^2 + \sum_{i=1}^n \left[(D \widehat{\otimes} 1)(1 \widehat{\otimes} f_i(z_i) \widehat{e}_i) \right. \\
 &\quad \left. + (1 \widehat{\otimes} f_i(z_i) \widehat{e}_i)(D \widehat{\otimes} 1) \right] \\
 &\quad + \sum_{i,j=1}^n (1 \widehat{\otimes} f_i(z_i) \widehat{e}_i)(1 \widehat{\otimes} f_j(z_j) \widehat{e}_j) \\
 &= D^2 \widehat{\otimes} 1 + \underbrace{\sum_{i=1}^n f_i(z_i)^2}_{\text{proper}} + \underbrace{\sum_{i=1}^n [D, f_i(z_i)] \widehat{\otimes} \widehat{e}_i}_{\text{bounded.}} \\
 \|[D, f_i(z_i)]\| &= |f'_i(z_i)| \underbrace{\|\nabla z_i\|}_{\leq 1+\varepsilon}.
 \end{aligned} \tag{2.10}$$

The norm here is the pointwise endomorphism norm. The commutator term is bounded. $\rightsquigarrow B$ is essentially self-adjoint and Fredholm. This uses the Callias Fredholm result. Throughout, $\text{Ind}(B)$ is the graded index: the dimension of the even kernel minus the dimension of the odd kernel.

Take $g_t = (1-t)\bar{g} + t g_o$, $0 \leq t \leq 1$; these metrics are uniformly equivalent to g_o . Use the homotopy $\theta_{t,i}(x) = (1-t)f_i(z_i(x)) + tx_i$ (potential of B). For sufficiently large $|x_i|$, both terms have the sign of x_i and grow at least linearly in $|x_i|$. Thus $\sum_i \theta_{t,i}^2$ tends to infinity uniformly in t , and all gradients are uniformly bounded. By homotopy invariance of the Fredholm index, $\text{Ind}(B) = \text{Ind}(B_o)$.

$$B_o = \sum_{i=1}^n \frac{\partial}{\partial x_i} e_i \widehat{\otimes} 1 + \sum_{i=1}^n 1 \widehat{\otimes} x_i \widehat{e}_i.$$

This operator acts on $L^2(\mathbb{R}^n, S(\mathbb{E}^n \oplus \mathbb{E}^n))$, and

$$B_o^2 = - \sum_{i=1}^n \frac{\partial^2}{\partial x_i^2} + \sum_{i=1}^n x_i^2 + \underbrace{\sum_{i=1}^n e_i \widehat{\otimes} e_i}_{\substack{\{e_i \widehat{\otimes} e_i\}_{1 \leq i \leq n} \\ \text{is a commuting family of symmetries}}}.$$

$$(e_i \widehat{\otimes} e_i)^2 = 1, \\ (e_i \widehat{\otimes} e_i)(e_j \widehat{\otimes} e_j) = (e_j \widehat{\otimes} e_j)(e_i \widehat{\otimes} e_i) \quad i \neq j.$$

For $i \neq j$, these symmetries commute.

Let Λ be the set of all maps $\lambda : \{1, \dots, n\} \longrightarrow \{1, -1\}$. For each $\lambda \in \Lambda$, define V_λ by

$$\underbrace{(e_i \widehat{\otimes} e_i)v}_{\text{acts as multiplication.}} = \underbrace{\lambda(i)}_{\text{Symmetry.}} v \quad \forall i.$$

There are 2^n maps λ . $\lambda(i) \in \{-1, 1\}$. The family $\{e_i \widehat{\otimes} e_i\}_{1 \leq i \leq n}$ decomposes $S(\mathbb{E}^n \oplus \mathbb{E}^n)$ into 2^n subspaces V_λ . Clifford multiplication by e_j interchanges the eigenspaces whose j -th signs differ, leaving the other signs unchanged. Thus all ranks are equal; since $\text{rank } S(\mathbb{E}^n \oplus \mathbb{E}^n) = 2^n$, each V_λ has rank 1.

On V_λ ,

$$B_o^2 = \underbrace{- \sum_{i=1}^n \frac{\partial^2}{\partial x_i^2} + \sum_{i=1}^n x_i^2}_{n\text{-dimensional harmonic oscillator}} + \sum_{i=1}^n \lambda(i).$$

In one dimension, $-\frac{d^2}{dx^2} + x^2$ has eigenvalues $O = \{1, 3, 5, \dots\}$. In n dimensions, the set of eigenvalues is $O_n := \{k_1 + k_2 + \dots + k_n : k_i \in O\}$. The smallest eigenvalue of the n -dimensional harmonic oscillator is $1 + 1 + \dots + 1 = n$. Hence the smallest eigenvalue of B_o^2 on V_λ is

$$n + \sum_{i=1}^n \lambda(i).$$

Since $\lambda(i) \in \{-1, 1\}$,

$$\sum_{i=1}^n \lambda(i) \geq -n,$$

with equality when $\lambda(i) = -1$ for every i .

If λ is not this constant map, at least one -1 is replaced by 1, and hence

$$\sum_{i=1}^n \lambda(i) \geq -n + 2.$$

Thus the smallest eigenvalue is at least $n + (-n) + 2 = 2$. For every $\lambda \neq -1$, B_o^2 has no 0 eigenvalue on $L^2(\mathbb{R}^n, V_\lambda)$.

For $\lambda \equiv -1$, $\sum_{i=1}^n \lambda(i) = -n$. Therefore,

$$B_o^2 = - \sum_{i=1}^n \frac{\partial^2}{\partial x_i^2} + \sum_{i=1}^n x_i^2 - n.$$

Its kernel is

$$\ker B_o^2 = \left\{ \exp\left(-\frac{x_1^2 + \cdots + x_n^2}{2}\right) v : v \in \underline{V_\lambda} \right\}.$$

Since $\text{rank } V_\lambda = 1$, $\ker B_o^2$ is one-dimensional. In the standard exterior-algebra model of the graded spinor module, the common -1 eigenspace consists of scalar forms, so V_λ is even. Self-adjointness gives $\ker B_o = \ker B_o^2$. Hence

$$\text{Ind}(B_o) = \dim(\text{even kernel}) - \dim(\text{odd kernel}) = 1 - 0 = 1 = \text{Ind}(B).$$

The full operator is self-adjoint; the graded index is nonzero.

Step 5. Lichnerowicz's formula: $D^2 = \nabla^* \nabla + \frac{\text{Sc}_{\bar{g}}}{4}$. There is no auxiliary curvature term because $\mathbb{B}^n$ is flat. The operator inequalities below are quadratic-form inequalities, initially on compactly supported smooth sections; $\|v\|$ denotes the L^2 -norm.

For $v \in C_c^\infty(\mathbb{R}^n, E)$,

$$Dv = \sum_{i=1}^n w_i \nabla_{w_i} v.$$

By Cauchy–Schwarz,

$$\|Dv\|^2 \leq n \|\nabla v\|^2, \quad \|\nabla v\|^2 \geq \frac{1}{n} \|Dv\|^2.$$

$$\begin{aligned} \|Dv\|^2 &= \|\nabla v\|^2 + \int_{\mathbb{R}^n} \frac{\text{Sc}_{\bar{g}}}{4} |v|^2 \\ &\geq \frac{1}{n} \|Dv\|^2 + \int_{\mathbb{R}^n} \frac{\text{Sc}_{\bar{g}}}{4} |v|^2. \end{aligned}$$

$$\Rightarrow \frac{n-1}{n} \|Dv\|^2 \geq \int_{\mathbb{R}^n} \frac{\text{Sc}_{\bar{g}}}{4} |v|^2 \quad \rightsquigarrow \quad D^2 \geq \frac{\text{Sc}_{\bar{g}}}{4} \frac{n}{n-1}.$$

By (2.10),

$$\begin{aligned}
 B^2 &\geq D^2 + \sum_{i=1}^n f_i(z_i)^2 - \sum_{i=1}^n \|[D, f_i(z_i)]\| \\
 &\geq \frac{\text{Sc}_{\bar{g}}}{4} \cdot \frac{n}{n-1} + \sum_{i=1}^n f_i(z_i)^2 - \sum_{i=1}^n f'_i(z_i) \|\nabla z_i\| \\
 &\geq \frac{\text{Sc}_{\bar{g}}}{4} \cdot \frac{n}{n-1} + \sum_{i=1}^n f_i(z_i)^2 - \sum_{i=1}^n f'_i(z_i)(1 + \varepsilon).
 \end{aligned}$$

Divide $\mathbb{R}^n$ into two cases.

The scalar-curvature lower bound is known only on the original cube: $\text{Sc}_g \geq n(n-1)$ on I^n .

Case 1. On I^n , $\bar{g} = g$, and $\text{Sc}_{\bar{g}} = \text{Sc}_g \geq n(n-1)$, $\frac{\text{Sc}_{\bar{g}}}{4} \frac{n}{n-1} \geq \frac{n^2}{4}$.

Outside the cube, the extended metric may have negative scalar curvature.

Recall $f'_i(z_i) = \xi_i(z_i, f_i(z_i))$, $(1 + \varepsilon)f'_i(z_i) \leq f_i(z_i)^2 + r_i^2 \Rightarrow f_i(z_i)^2 - (1 + \varepsilon)f'_i(z_i) \geq -r_i^2$.

$$\begin{aligned}
 B^2 &\geq \frac{n^2}{4} + \sum_{i=1}^n \left(f_i(z_i)^2 - (1 + \varepsilon)f'_i(z_i) \right) \\
 &\geq \frac{n^2}{4} - \sum_{i=1}^n \underbrace{r_i^2}_{\frac{\pi^2(1 + \varepsilon)^2}{(\ell_i - 2\varepsilon)^2}} > \delta.
 \end{aligned}$$

Case 2. Outside the cube,

$$x \notin I^n \Rightarrow \text{for some } i, \quad |z_i(x)| \geq \frac{\ell_i}{2} - \varepsilon.$$

$$\text{Without loss of generality, } |z_1| \geq \underbrace{\frac{\ell_1}{2} - \varepsilon}_{> \bar{\ell}_1/2}.$$

Outside: positivity comes from the potential.

On $|x| \leq \bar{\ell}_1/2$, $f_1(x) = r_1 \tan\left(\frac{r_1 x}{1 + \varepsilon}\right)$. Since f_1 is nondecreasing, for $|x| \geq \ell_1/2 - \varepsilon$ we have

$$\begin{aligned}
 |f_1(x)| &\geq \underbrace{r_1}_{\frac{\pi(1 + \varepsilon)}{\ell_1 - 2\varepsilon}} \tan\left(\frac{r_1 \bar{\ell}_1}{2(1 + \varepsilon)}\right)
 \end{aligned}$$

Also, $|f'_1(z_1)| \leq \varepsilon$. $\text{Sc}_{\bar{g}} \geq -\sigma$. For $i \geq 2$, $f_i(z_i)^2 - (1 + \varepsilon)f'_i(z_i) \geq -r_i^2$.

$$\begin{aligned} B^2 &\geq -\frac{n\sigma}{4(n-1)} + f_1(z_1)^2 - \varepsilon(1 + \varepsilon) - \sum_{i=2}^n r_i^2 \\ &\geq -\frac{n\sigma}{4(n-1)} + \frac{\pi^2(1 + \varepsilon)^2}{(\ell_1 - 2\varepsilon)^2} \tan^2\left(\frac{\pi\bar{\ell}_1}{2(\ell_1 - 2\varepsilon)}\right) - \varepsilon(1 + \varepsilon) - \sum_{i=2}^n r_i^2 \\ &> \delta. \end{aligned}$$

Thus $\|Bv\|^2 \geq \delta\|v\|^2$ on the domain of B , by closure from the smooth core. Since B is self-adjoint, B is invertible. $(\text{Ran } B)^\perp = \ker B = 0$, so the range is dense and closed. An invertible graded Fredholm operator has no kernel, so $\text{Ind}(B) = 0$. ∇

Part II: strict inequality

To prove strictness under the dihedral angle assumption, we need to exclude

$$\underbrace{\sum_{i=1}^n \frac{1}{\ell_i^2}}_{\text{assume to the contrary.}} = \frac{n^2}{4\pi^2}.$$

Step 1. Let $E := S(TI^n \oplus \mathbb{E}^n)$ be the bundle over I^n , and set

$$B := D \widehat{\otimes} 1 - \sum_{i=1}^n 1 \widehat{\otimes} \underbrace{f_i(z_i)}_{\substack{\text{to be specified} \\ \text{later}}} \widehat{e}_i.$$

Here D and the flat auxiliary connection are as in Part I, restricted to I^n . The minus sign in this part matches the inward-normal boundary condition and the negative tangent functions below. For now, assume the compositions $f_i(z_i)$ are smooth on the closed cube.

Boundary: the tangent term is very large near the faces; the boundary argument ultimately forces a kernel section to vanish on ∂I^n .

Choose a manifold with corners $M \subset \mathbb{R}^n$ and an orientation-preserving corner diffeomorphism $\Phi : I^n \rightarrow M$ as follows. For sufficiently large $R > 1$, set

$$M = \bigcap_{i=1}^n (\overline{B}(\widehat{e}_i, R) \cap \overline{B}(-\widehat{e}_i, R)), \quad \Phi(x) = \frac{(R^2 - 1)(2x_1 - 1, \dots, 2x_n - 1)}{\sqrt{1 + (R^2 - 1) \sum_{j=1}^n (2x_j - 1)^2 + 1}}.$$

Here $\overline{B}(y, R)$ is the closed Euclidean ball of radius R centered at y . The inverse has coordinates $x_i = \frac{1}{2} + y_i/(R^2 - 1 - |y|^2)$, so this is a smooth diffeomorphism. The faces $\Phi(\partial_{i\pm})$ lie on the spheres centered at $\mp \widehat{e}_i$.

(1) Φ preserves the corner structures.

$$\begin{array}{ccc} \text{codimensions 1, 2} & & \text{codimensions 1, 2} \\ \downarrow & & \downarrow \\ \text{face of } I^n & \longrightarrow & \text{face of } M \end{array}$$

(2) At each codimension-2 face,

$$0 < \underbrace{\theta(x)}_{\text{dihedral angle of } (I^n, g)} < \underbrace{\theta_M(\Phi(x))}_{\text{of } M} < \pi$$

Indeed, every dihedral angle of M is $\pi - \arccos(1 - R^{-2})$, which tends to π as R tends to infinity. The angles of (I^n, g) have maximum strictly less than π , so such an R exists.

↪ Use index theory for manifolds with corners.

(3) On any closed codimension-one face F of I^n , use the inward normal to $\Phi(F)$. Its components satisfy

$$\begin{aligned} \langle \widehat{\nu}, \widehat{e}_i \rangle &< 0 \quad \text{on } \Phi(F \cap \partial_{i+}), \\ \langle \widehat{\nu}, \widehat{e}_i \rangle &> 0 \quad \text{on } \Phi(F \cap \partial_{i-}). \end{aligned}$$

At $y \in \Phi(\partial_{j\pm})$, this normal is $-(y \pm \widehat{e}_j)/R$. The formula for Φ shows the asserted signs for every incident coordinate, including at corners.

On each codimension-one face $F \subset I^n$, the boundary condition $\mathfrak{B}$ is given by

$$(\nu_x \widehat{\otimes} \widehat{\nu}_{\Phi(x)})\varphi(x) = -\varphi(x), \quad \Phi(F) \subset M.$$

Here ν_x is the inward unit normal to F . The corner index theorem applies: both tangent bundles are trivial, so Φ is spin. It has degree 1, and M is diffeomorphic to the cube, so its Euler characteristic is 1. The smooth potential is a bounded self-adjoint zeroth-order perturbation, so the self-adjoint domain is unchanged. It is compact as a map $H^1 \rightarrow L^2$, so the Fredholm index is unchanged. Hence:

- (1) B with $\mathfrak{B}$ is self-adjoint and Fredholm, with domain $H^1(I^n, E; \mathfrak{B})$.
- (2) $\text{Ind}(B) = 1$. Therefore, there exists a **nonzero** $\varphi \in H^1(I^n, E; \mathfrak{B})$ such that $B\varphi = 0$.

Step 2. Bochner formula. Let ψ be a smooth section of E over I^n that vanishes near all codimension-2 faces and satisfies $\mathfrak{B}$ on every codimension-one face.

$$\begin{aligned} \int_{I^n} \langle B\psi, B\psi \rangle &= \int_{I^n} |D\psi|^2 \\ &+ \int_{I^n} \left\langle \left(\sum_{i=1}^n f_i(z_i)^2 - \sum_{i=1}^n \underbrace{[D, f_i(z_i)] \widehat{\otimes} \widehat{e}_i}_{\text{commutator}} \right) \psi, \psi \right\rangle \\ &- \underbrace{\int_{\partial I^n}}_{\text{boundary}} \sum_{i=1}^n \langle f_i(z_i) (\nu \widehat{\otimes} \widehat{e}_i) \psi, \psi \rangle. \end{aligned}$$

Part I: $|D\psi|^2 \leq n|\nabla\psi|^2 \longrightarrow$ discards information in the equality case.

Penrose operator. $P_V\psi := \nabla_V\psi + \frac{1}{n}V \cdot D\psi, \quad V \in TI^n.$

$$|\nabla\psi|^2 = \underbrace{|P\psi|^2}_{\geq 0} + \underbrace{\frac{1}{n}|D\psi|^2}_{\geq 0}. \quad \rightsquigarrow \quad |D\psi|^2 \leq n|\nabla\psi|^2$$

↘ Records the equality condition in the Lichnerowicz formula.

The Lichnerowicz formula gives

$$\begin{aligned} \int_{I^n} |D\psi|^2 &= \int_{I^n} \left(\frac{\text{Sc}_g}{4} |\psi|^2 + |\nabla\psi|^2 \right) + \int_{\partial I^n} \langle (\nu \cdot D + \nabla_\nu)\psi, \psi \rangle \\ \underbrace{\left(1 - \frac{1}{n} \right)}_{\frac{n-1}{n}} \int_{I^n} |D\psi|^2 &= \int_{I^n} \left(\frac{\text{Sc}_g}{4} |\psi|^2 + |P\psi|^2 \right) + \int_{\partial I^n} \langle (\nu \cdot D + \nabla_\nu)\psi, \psi \rangle. \\ \Rightarrow \int_{I^n} |D\psi|^2 &= \frac{n}{n-1} \int_{I^n} \left(\frac{\text{Sc}_g}{4} |\psi|^2 + |P\psi|^2 \right) + \frac{n}{n-1} \int_{\partial I^n} \langle (\nu \cdot D + \nabla_\nu)\psi, \psi \rangle \end{aligned}$$

On each codimension-one face of I^n ,

$$\nu \cdot D + \nabla_\nu = \sum_{j=1}^{n-1} \nu \cdot e_j \cdot \nabla_{e_j},$$

where $\{e_j\}_{j=1}^{n-1}$ is a tangent orthonormal frame.

Define

$$\begin{aligned} \mathcal{H} := -\frac{1}{2} \Big[& (\nu \cdot D + \nabla_\nu)(\nu \widehat{\otimes} \widehat{\nu}) \\ & + (\nu \widehat{\otimes} \widehat{\nu})(\nu \cdot D + \nabla_\nu) \Big]. \end{aligned}$$

The first-order terms cancel because the tangential Clifford symbols anticommute with $\nu \widehat{\otimes} \widehat{\nu}$. Thus $\mathcal{H}$ is a bounded endomorphism. The boundary condition gives

$$\langle (\nu \cdot D + \nabla_\nu)\psi, \psi \rangle = \langle \mathcal{H}\psi, \psi \rangle.$$

Integration by parts on a manifold with boundary shows that $\nu \cdot D + \nabla_\nu$ is not automatically nonnegative.

At a point of a face of M , decompose

$$\widehat{e}_i = \underbrace{a_i}_{\langle \widehat{\nu}, \widehat{e}_i \rangle} \widehat{\nu} + \underbrace{\widehat{w}_i}_{\perp \widehat{\nu}}.$$

$$\langle (\nu \widehat{\otimes} \widehat{e}_i) \psi, \psi \rangle = a_i \langle (\nu \widehat{\otimes} \widehat{\nu}) \psi, \psi \rangle + \underbrace{\langle (\nu \widehat{\otimes} \widehat{w}_i) \psi, \psi \rangle}_{\substack{\text{anticommutes} \\ \swarrow = 0}}$$

The boundary condition makes the first term $-a_i |\psi|^2$. The second vanishes because its operator interchanges the two eigenspaces of the self-adjoint involution $\nu \widehat{\otimes} \widehat{\nu}$.

The following identity holds:

$$\begin{aligned} \int_{I^n} \langle B\psi, B\psi \rangle &= \frac{n}{n-1} \int_{I^n} |P\psi|^2 \\ &+ \int_{I^n} \left\langle \left(\frac{n}{n-1} \frac{\text{Sc}_g}{4} + \sum_{i=1}^n f_i(z_i)^2 - \sum_{i=1}^n [D, f_i(z_i)] \widehat{\otimes} \widehat{e}_i \right) \psi, \psi \right\rangle \\ &+ \frac{n}{n-1} \int_{\partial I^n} \langle \mathcal{H}\psi, \psi \rangle \\ &+ \int_{\partial I^n} \left(\sum_{i=1}^n \langle \widehat{\nu}, \widehat{e}_i \rangle f_i(z_i) \right) |\psi|^2 \end{aligned} \quad (\star)$$

This holds for every $\psi \in H^1(I^n, E; \mathfrak{B})$. Indeed, smooth sections satisfying the boundary condition and vanishing near the codimension-two faces are dense in this H^1 -space. Both sides are H^1 -continuous, including the boundary terms by the trace theorem.

Step 3. Define $f_i(x) = -\frac{\pi}{\ell_i} \tan\left(\frac{\pi x}{\ell_i}\right)$ $x \in (-\ell_i/2, \ell_i/2)$.

$$\begin{aligned} |f'_i(x)| &= \left| -\frac{\pi}{\ell_i} \frac{\pi}{\ell_i} \sec^2\left(\frac{\pi x}{\ell_i}\right) \right| \\ &= \frac{\pi^2}{\ell_i^2} \underbrace{\sec^2\left(\frac{\pi x}{\ell_i}\right)}_{1 + \tan^2\left(\frac{\pi x}{\ell_i}\right)} \\ f_i^2(x) &= \frac{\pi^2}{\ell_i^2} \tan^2\left(\frac{\pi x}{\ell_i}\right). \end{aligned}$$

$$\rightsquigarrow |f'_i(x)| = \frac{\pi^2}{\ell_i^2} + f_i^2(x) \quad \Rightarrow \quad \frac{\pi^2}{\ell_i^2} + f_i^2 - |f'_i| = 0, \quad f_i(x) \longrightarrow \mp\infty, \quad x \longrightarrow \pm\ell_i/2.$$

Recall that $\ell_i = \text{dist}_g(\partial_{i-}, \partial_{i+})$. Start with the 1-Lipschitz function

$$x \longmapsto \min\{\ell_i, \text{dist}_g(x, \partial_{i-})\} - \frac{\ell_i}{2}.$$

$$\begin{aligned} \text{on } \partial_{i-}, \quad & \rightsquigarrow 0 \quad \Rightarrow \quad -\frac{\ell_i}{2} \\ \text{on } \partial_{i+}, \quad & \rightsquigarrow \ell_i \quad \Rightarrow \quad \frac{\ell_i}{2}. \end{aligned}$$

For a choice that is strictly between these endpoint values away from the two faces, set

$$\widetilde{z}_i(x) = \frac{\ell_i \operatorname{dist}_g(x, \partial_{i-}) - \operatorname{dist}_g(x, \partial_{i+})}{2 \operatorname{dist}_g(x, \partial_{i-}) + \operatorname{dist}_g(x, \partial_{i+})}.$$

The denominator is at least ℓ_i , by the triangle inequality. Since both distance functions are 1-Lipschitz, differentiation almost everywhere gives

$$\|\nabla \widetilde{z}_i(x)\| \leq \frac{\ell_i}{\operatorname{dist}_g(x, \partial_{i-}) + \operatorname{dist}_g(x, \partial_{i+})} \leq 1.$$

Thus $\widetilde{z}_i$ is 1-Lipschitz, equals $\pm \ell_i/2$ on $\partial_{i\pm}$, and satisfies $|\widetilde{z}_i| < \ell_i/2$ elsewhere.

On each closed face, write $a_i = \langle \widehat{\nu}, \widehat{e}_i \rangle$, using that face's normal. For $0 < t < 1$, consider $(1-t)\widetilde{z}_i$. On $\Phi(\partial_{i-})$, $a_i > 0$, and for $x \in \partial_{i-}$,

$$\begin{aligned} (1-t)\widetilde{z}_i(x) &= -(1-t)\frac{\ell_i}{2} \longrightarrow -\frac{\ell_i}{2}, \quad f_i((1-t)\widetilde{z}_i(x)) \longrightarrow +\infty, \\ \Rightarrow \quad a_i f_i((1-t)\widetilde{z}_i(x)) &\longrightarrow +\infty \quad (t \downarrow 0, x \in \partial_{i-}). \end{aligned}$$

On $\Phi(\partial_{i+})$, $a_i < 0$, and for $x \in \partial_{i+}$,

$$\begin{aligned} (1-t)\widetilde{z}_i(x) &= (1-t)\frac{\ell_i}{2} \longrightarrow \frac{\ell_i}{2}, \quad f_i((1-t)\widetilde{z}_i(x)) \longrightarrow -\infty, \\ \Rightarrow \quad a_i f_i((1-t)\widetilde{z}_i(x)) &\longrightarrow +\infty \quad (t \downarrow 0, x \in \partial_{i+}). \end{aligned}$$

Claim 2.3. $\sum_{i=1}^n a_i(x) f_i((1-t)\widetilde{z}_i(x)) \longrightarrow +\infty \quad (t \downarrow 0)$ uniformly on each closed codimension-one face.

Fix $F = \partial_{j\pm}$. Its j -th term tends uniformly to $+\infty$, since $\widetilde{z}_j = \pm \ell_j/2$ there and $|a_j|$ has a positive minimum. For $i \neq j$, near $F \cap \partial_{i\pm}$, the incident-face sign condition makes $a_i f_i((1-t)\widetilde{z}_i)$ nonnegative for sufficiently small t . Away from these intersections, $\widetilde{z}_i$ stays uniformly away from $\pm \ell_i/2$, so this term is uniformly bounded. Thus the remaining terms are uniformly bounded below, and the j -th term dominates them. Compactness of F , and the finite number of faces, give the claim.

Step 4. First fix a sufficiently small $t > 0$ and replace $\widetilde{z}_i$ by $(1-t)\widetilde{z}_i$. The claim gives a strict boundary margin. This contraction also leaves a Lipschitz constant at most $1-t$ and keeps the range away from the poles. Smooth sufficiently closely, with an increase in Lipschitz constant smaller than t , to obtain smooth 1-Lipschitz functions z_i . At corners, use local inward translations before convolution and then a partition of unity; the increase in Lipschitz constant can still be made arbitrarily small. The strict range and boundary

inequalities persist: $|z_i(x)| < \ell_i/2$ for every $x \in I^n$, and

$$\begin{aligned} &\text{For every } x \in \partial_{j\pm}, \quad 1 \leq j \leq n, \\ &\sum_{i=1}^n \langle \widehat{v}, \widehat{e}_i \rangle f_i(z_i(x)) > \frac{n}{n-1} \|\mathcal{H}\|. \end{aligned} \quad \leftarrow \text{perturbation near the boundary}$$

Since $\text{Sc}_g \geq n(n-1)$,

$$\frac{n}{n-1} \frac{\text{Sc}_g}{4} \geq \frac{n^2}{4}.$$

By 1-Lipschitzness, $\|\nabla z_i\| \leq 1$, so

$$\begin{aligned} \|[D, f_i(z_i)]\| &= |f'_i(z_i)| \|\nabla z_i\| \leq |f'_i(z_i)|, \\ -[D, f_i(z_i)] \widehat{\otimes} \widehat{e}_i &\geq -\|[D, f_i(z_i)]\| \geq -|f'_i(z_i)|. \end{aligned}$$

Using $f_i(z_i)^2 - |f'_i(z_i)| = -\pi^2/\ell_i^2$, we obtain

$$\begin{aligned} &\frac{n}{n-1} \frac{\text{Sc}_g}{4} + \sum_{i=1}^n f_i(z_i)^2 - \sum_{i=1}^n [D, f_i(z_i)] \widehat{\otimes} \widehat{e}_i \\ &\geq \frac{n^2}{4} + \sum_{i=1}^n f_i(z_i)^2 - \sum_{i=1}^n |f'_i(z_i)| \\ &= \frac{n^2}{4} - \underbrace{\sum_{i=1}^n \frac{\pi^2}{\ell_i^2}}_{=\frac{n^2}{4}} = 0. \end{aligned}$$

Let $\varphi \in H^1(I^n, E; \mathfrak{B})$ be the nonzero section from Step 1 such that $B\varphi = 0$. By $(\star)$, $\int_{I^n} \langle B\varphi, B\varphi \rangle = 0$.

$$\begin{aligned} \frac{n}{n-1} \langle \mathcal{H}\varphi, \varphi \rangle + \left(\sum_{i=1}^n \langle \widehat{v}, \widehat{e}_i \rangle f_i(z_i) \right) |\varphi|^2 &\geq \left(\sum_{i=1}^n \langle \widehat{v}, \widehat{e}_i \rangle f_i(z_i) - \frac{n}{n-1} \|\mathcal{H}\| \right) |\varphi|^2 \\ &\geq 0. \end{aligned}$$

The coefficient in parentheses is strictly positive on ∂I^n . Since the interior terms and the combined boundary term in $(\star)$ are nonnegative, equality gives

$$\int_{I^n} |P\varphi|^2 = 0 \quad \Rightarrow \quad P\varphi = 0 \quad \text{almost everywhere on } I^n.$$

$\varphi = 0$ almost everywhere on ∂I^n .

Since $B\varphi = 0$ and $P\varphi = 0$,

$$\begin{aligned} \nabla_V \varphi + \frac{1}{n} V \cdot D\varphi &= 0 \quad \Rightarrow \quad \nabla_V \varphi = -\frac{1}{n} V \cdot D\varphi, \\ D\varphi - \sum_{i=1}^n f_i(z_i) \widehat{e}_i \varphi &= 0 \quad \Rightarrow \quad D\varphi = \sum_{i=1}^n f_i(z_i) \widehat{e}_i \varphi, \\ \nabla_V \varphi &= -\frac{1}{n} V \cdot \left(\sum_{i=1}^n f_i(z_i) \widehat{e}_i \varphi \right) \\ &= -\frac{1}{n} \left(\sum_{i=1}^n V \widehat{\otimes} f_i(z_i) \widehat{e}_i \right) \varphi, \quad \text{for every } V \in TI^n. \end{aligned}$$

In a smooth local trivialization, this expresses each weak first derivative of φ as a smooth matrix times φ . Iteration gives H^k -regularity for every k , hence smoothness up to the boundary by Sobolev embedding. Its zero trace therefore gives $\varphi(y_0) = 0$ for any $y_0 \in \partial I^n$.

Let $x \in I^n$. Choose a smooth path γ_x connecting x to y_0 . Let $V = \dot{\gamma}_x$.

$$\nabla_{\dot{\gamma}_x} \varphi = -\frac{1}{n} \left(\sum_{i=1}^n \dot{\gamma}_x \widehat{\otimes} f_i(z_i) \widehat{e}_i \right) \varphi.$$

By uniqueness, $\varphi(y_0) = 0$ implies that $\varphi = 0$ along γ_x . Hence $\varphi(x) = 0$. Since $x \in I^n$ was arbitrary, $\varphi = 0$ on I^n , a contradiction. $\square$

3 Gromov's simplicial volume vanishing conjecture for positive scalar curvature and fundamental group decay

Gromov's conjecture ([MY26, (1.2)]). *Let M be closed and oriented. If M admits positive scalar curvature (PSC), then $\| [M] \|_{\ell^1} = 0$.*

Assumptions. The universal cover $\tilde{M}$ is spin, and the fundamental group has sufficiently small decay rate.

3.1 Scalar curvature and simplicial volume

3.1.1 Scalar curvature

At a point $p \in M$, scalar curvature is characterized by

$$\frac{\text{Vol}(B_M(p, r))}{\text{Vol}(B_{\mathbb{R}^m}(r))} = 1 - \frac{\text{Scal}(p)}{6(m+2)} r^2 + o(r^2), \quad \frac{o(r^2)}{r^2} \rightarrow 0 \quad (r \rightarrow 0).$$

PSC: very small balls have smaller volume than Euclidean balls. Moreover,

$$\text{Scal}(p) = 6(m+2) \lim_{r \rightarrow 0} \frac{1 - \text{Vol}(B_M(p, r)) / \text{Vol}(B_{\mathbb{R}^m}(0, r))}{r^2}.$$

Thus, if $\text{Scal}(p) > 0$, then $\text{Vol}(B_M(p, r)) < \text{Vol}(B_{\mathbb{R}^m}(0, r))$ for sufficiently small r .

The normalization is $\text{Scal}(S^m) = m(m-1) \rightsquigarrow \text{Scal} \geq m(m-1)$.

Theorem 3.1 (Lichnerowicz [MY26, (1.1)]). *If M is closed and spin and M admits PSC, then $\widehat{A}(M) = 0$.*

3.1.2 Simplicial volume

Let M be a closed oriented m -manifold. A singular chain $c = \sum_i a_i \sigma_i$ represents the fundamental class $[M] \in H_m(M; \mathbb{R})$ if $[c] = [M]$, where $a_i \in \mathbb{R}$ and $\sigma_i: \Delta^m \rightarrow M$ is a continuous singular simplex. One may triangulate the manifold. Here

$$\Delta^m = \left\{ (t_0, \dots, t_m) \in \mathbb{R}^{m+1} : t_j \geq 0, \sum_{j=0}^m t_j = 1 \right\}.$$

Definition 3.2 (Simplicial volume [MY26, (1.3)]). The simplicial volume of M is

$$\begin{aligned} \|M\| &= \|[M]\|_{\ell^1} = \inf_{[c]=[M]} \sum_i |a_i| \\ &= \inf \left\{ \sum_i |a_i| : c = \sum_i a_i \sigma_i, \partial c = 0, [c] = [M] \right\}, \end{aligned}$$

where $\|c\|_{\ell^1} = \sum_i |a_i|$. It measures how efficiently the fundamental class can be built from singular simplices.

Intuition. Let $d \gg 1$ be a positive integer. Define $\sigma_d: [0, 1] \rightarrow S^1$ by $\sigma_d(t) = e^{2\pi i d t}$. Then $\sigma_d(0) = 1 = \sigma_d(1)$ and $\partial \sigma_d = \sigma_d(1) - \sigma_d(0) = 0$. Thus σ_d is a singular 1-cycle which winds d times

around a circle. Since $[\sigma_d] = d[S^1]$, we have $\frac{1}{d}[\sigma_d] = [S^1]$ and $\|\frac{1}{d}\sigma_d\|_{\ell^1} = \frac{1}{d}$. Therefore $\|S^1\| \leq \frac{1}{d}$ for every positive integer d , so $\|S^1\| = 0$, but $[S^1] \neq 0$.

More generally, for S^n , $n \geq 1$, use self-maps $f_d: S^n \rightarrow S^n$ of degree $d \geq 1$. Then $\|f_{d*}[S^n]\|_{\ell^1} \leq \|[S^n]\|_{\ell^1}$. Since $f_{d*}[S^n] = d[S^n]$, it follows that $d\|S^n\| \leq \|S^n\|$, and hence $\|S^n\| = 0$.

Example 3.3 (Hyperbolic surfaces). Let Σ_g be a closed oriented surface of genus $g \geq 2$. Equip Σ_g with a hyperbolic metric of curvature -1 , so $K = -1$. Let ω be the area form. Then $\text{Area}(\Sigma_g) = \int_{\Sigma_g} \omega$ and $K\omega = -\omega$. By Gauss–Bonnet, $\int_{\Sigma_g} K\omega = 2\pi\chi(\Sigma_g)$. Hence

$$\int_{\Sigma_g} K\omega = - \int_{\Sigma_g} \omega = -\text{Area}(\Sigma_g) = 2\pi\chi(\Sigma_g),$$

and therefore $\text{Area}(\Sigma_g) = -2\pi\chi(\Sigma_g)$.

Let $c = \sum_i a_i \sigma_i$ be a singular fundamental cycle, and let $p: \mathbb{H}^2 \rightarrow \Sigma_g$ be the universal covering map. The metric on Σ_g lifts to a hyperbolic metric on $\mathbb{H}^2$.

For $\sigma_i: \Delta^2 \rightarrow \Sigma_g$, choose a lift $\tilde{\sigma}_i: \Delta^2 \rightarrow \mathbb{H}^2$ such that $p \circ \tilde{\sigma}_i = \sigma_i$.

Idea: replace $\tilde{\sigma}_i$ by the geodesic triangle with the same vertices and project back to Σ_g . This gives the straightening of σ_i . Set $\text{st}(c) = \sum_i a_i \text{st}(\sigma_i)$. It represents the same fundamental class: $[\text{st}(c)] = [c] = [\Sigma_g]$. Thus

$$\begin{aligned} \text{Area}(\Sigma_g) &= \langle \omega, \text{st}(c) \rangle \\ &= \left| \sum_i a_i \int_{\text{st}(\sigma_i)} \omega \right| \\ &\leq \sum_i |a_i| \left| \int_{\text{st}(\sigma_i)} \omega \right| \\ &\leq \pi \sum_i |a_i|. \end{aligned}$$

Therefore $\sum_i |a_i| \geq \text{Area}(\Sigma_g)/\pi$. Taking the infimum, $\|\Sigma_g\| \geq \text{Area}(\Sigma_g)/\pi = -2\chi(\Sigma_g) > 0$.

(Gromov) If M is a closed hyperbolic manifold, then $\|M\| = \text{Vol}(M)/v_m > 0$, where v_m is the volume of a regular ideal simplex in $\mathbb{H}^m$. For $m \geq 3$, a homotopy equivalence between closed hyperbolic m -manifolds is homotopic to an isometry.

(Gromov) If M has amenable fundamental group, then $\|M\| = 0$.

The straightening method gives the non-vanishing of simplicial volume (Thurston).

(Gromov) PSC conjecture on simplicial volume: PSC forces the simplicial volume $\|M\|$ to vanish.

Recall: S^n admits PSC and $\|S^n\| = 0$. A closed hyperbolic surface satisfies $\|\Sigma_g\| > 0$, and its hyperbolic metric has negative scalar curvature.

3.2 The Ma–Yu theorem and fundamental group decay

Theorem 3.4 (Ma–Yu [MY26, Theorem 1.1 and the following discussion]). *There exists a universal constant $C > 0$, dependent only on the dimension m , such that if M is an m -dimensional closed oriented Riemannian manifold whose universal cover $\tilde{M}$ is spin and which satisfies $\text{Scal} \geq C$ and $\text{Ric} \geq -1$, then $\|M\| = 0$.*

The lower bound $\text{Ric} \geq -1$ means that $\text{Ric}_p(v, v) \geq -g_p(v, v) = -|v|^2$ for $p \in M$ and $v \in T_p M$. If v is a unit vector and $(v, e_2, \dots, e_m)$ is an orthonormal basis of $T_p M$, then $\text{Ric}_p(v, v) = K_p(v, e_2) + \dots + K_p(v, e_m) \geq -1$.

Definition 3.5 (Volume entropy [MY26, (1.6)]). The volume entropy is $h_g = \limsup_{R \rightarrow \infty} \frac{\ln \text{Vol}(B_{\widetilde{M}}(x, R))}{R}$.

If $\text{Vol}(B(x, R)) \sim R^m$, then $h = 0$. If $\text{Vol}(B(x, R)) \sim e^{aR}$, then $h = a$.

The Ricci curvature lower bound may be replaced by an upper bound 1 on h_g .

3.2.1 Fundamental group decay

Let $x_b \in \widetilde{M}$ and $\Gamma = \pi_1(M)$. Define $d_\Gamma(\gamma, \gamma') = d_{\widetilde{M}}(\gamma x_b, \gamma' x_b)$, which is inherited from the Riemannian metric. Equivalently, the corresponding length function is $\ell(\gamma) = d_{\widetilde{M}}(x_b, \gamma x_b)$, and $B_\Gamma(R) = B(1_\Gamma, R) = \{\gamma \in \Gamma : \ell(\gamma) \leq R\}$.

For $a \in C_c(\Gamma)$, the function $a : \Gamma \rightarrow \mathbb{C}$ has finite support, $a = \sum_{\gamma \in \Gamma} a(\gamma)\gamma$, and $\|a\|_{\ell^2(\Gamma)}^2 = \sum_{\gamma \in \Gamma} |a(\gamma)|^2$. If $(a * b)(\gamma) = \sum_{\gamma' \gamma'' = \gamma} a(\gamma')b(\gamma'')$, then $\|a\|_{C_r^*(\Gamma)} = \sup_{0 \neq b \in C_c(\Gamma)} \frac{\|a * b\|_2}{\|b\|_2}$.

Definition 3.6 (Exponential decay property). The fundamental group Γ is said to satisfy the exponential decay property at rate $d \geq 0$ if there exists a constant $C > 0$ such that, for every $R > 0$ and every $a \in C_c(\Gamma)$,

$$\|a\|_{C_r^*(\Gamma)}^2 \leq C e^{2dR} \sum_{\gamma \in \Gamma} |a(\gamma)|^2$$

whenever $\text{supp}(a) \subseteq B_\Gamma(R)$. Equivalently, $\|a\|_{C_r^*(\Gamma)} \leq \sqrt{C} e^{dR} \|a\|_2$.

Remark 3.7. Replace the exponential function by a polynomial function. This gives the classical property RD.

Define a convolution operator on $\ell^2(\Gamma)$ by $(\lambda(a)\xi)(x) = \sum_{\gamma \in \Gamma} a(\gamma)\xi(\gamma^{-1}x)$. Then $\|a\|_{C_r^*(\Gamma)} = \|\lambda(a)\|_{\mathcal{B}(\ell^2(\Gamma))}$. The ratio $\|a\|_{C_r^*(\Gamma)} / \|a\|_{\ell^2(\Gamma)}$ measures how large convolution by a can be compared with the size of its coefficients.

Definition 3.8 (Decay function and decay rate [MY26, (1.12)]). Define the decay function $\Psi_\Gamma(R)$ and the decay rate ψ_Γ by

$$\Psi_\Gamma(R) = \log \left(\sup_{\substack{0 \neq a \in C_c(\Gamma) \\ \text{supp}(a) \subseteq B(1_\Gamma, R)}} \frac{\|a\|_{C_r^*(\Gamma)}}{\|a\|_{\ell^2(\Gamma)}} \right), \quad \psi_\Gamma = \limsup_{R \rightarrow \infty} \frac{\Psi_\Gamma(R)}{R}.$$

Every $d > \psi_\Gamma$ is an admissible exponential decay rate.

Definition 3.9 (Property RD [MY26, (1.20)]). Property RD means that there exist constants $C > 0$ and $s > 0$ such that, for every $R > 0$ and every $a \in C_c(\Gamma)$ with $\text{supp}(a) \subseteq B(1_\Gamma, R)$, one has $\|a\|_{C_r^*(\Gamma)} \leq C(1 + R)^s \|a\|_{\ell^2(\Gamma)}$.

Therefore

$$\sup_{\substack{0 \neq a \in C_c(\Gamma) \\ \text{supp}(a) \subseteq B(1_\Gamma, R)}} \frac{\|a\|_{C_r^*(\Gamma)}}{\|a\|_{\ell^2(\Gamma)}} \leq C(1 + R)^s,$$

and

$$\frac{\Psi_\Gamma(R)}{R} \leq \frac{\log C}{R} + s \frac{\log(1+R)}{R}.$$

Since $\frac{\log C}{R} \rightarrow 0$ and $\frac{\log(1+R)}{R} \rightarrow 0$, we obtain $\psi_\Gamma \leq 0$.

If $a = \delta_{1_\Gamma}$, then $\|a\|_{C_r^*(\Gamma)} = \|a\|_{\ell^2(\Gamma)} = 1$. Thus $\Psi_\Gamma(R) \geq 0$, so $\psi_\Gamma \geq 0$. Consequently, $\psi_\Gamma = 0$.

By Cauchy–Schwarz, $\|a\|_{C_r^*(\Gamma)} \leq \|a\|_1 \leq |B_\Gamma(R)|^{1/2} \|a\|_2$. Hence $\psi_\Gamma \leq \limsup_{R \rightarrow \infty} \frac{\log |B_\Gamma(R)|}{2R}$. Because M is compact, orbit growth can be compared with volume growth: $|B_\Gamma(R)| \leq C \text{Vol}(B_{\tilde{M}}(x, R+C))$. Therefore $\psi_\Gamma \leq h_g/2$.

Theorem 3.10 (Main theorem [MY26, Theorem 1.2 and the preceding discussion]). *There exists a universal constant $C > 0$, depending only on the dimension m , such that if M is a closed oriented manifold of dimension m whose universal cover $\tilde{M}$ is spin, and M admits a Riemannian metric with scalar curvature $\text{Scal} \geq C$ and decay rate $\psi_\Gamma \leq 1$, then $\|M\| = 0$.*

The spinor bundle is $S_{\tilde{M}}$, and the Dirac operator is on $\tilde{M}$.

3.3 Dirac operators

Let $D = C_1 \frac{\partial}{\partial x_1} + \cdots + C_m \frac{\partial}{\partial x_m}$, where $C_i^2 = -1$ and $C_i C_j + C_j C_i = 0$ for $i \neq j$. Then $D^2 = -\sum_{i=1}^m \frac{\partial^2}{\partial x_i^2} = \Delta$, the Laplacian. This is a Dirac operator retaining the Clifford algebra structure.

Let $e_1, \dots, e_m$ be an orthonormal frame. The spin Dirac operator is $D = \sum_{j=1}^m c(e_j) \nabla_{e_j}$, where $c(e_j)$ denotes Clifford multiplication. It differentiates spinors in the directions e_j .

The Lichnerowicz formula is $D^2 = \nabla^* \nabla + \text{Scal}/4$, where $\nabla^* \nabla \geq 0$. If $\text{Scal} \geq k > 0$, then for every smooth compactly supported spinor s ,

$$\langle D^2 s, s \rangle = \|\nabla s\|^2 + \frac{1}{4} \int_{\tilde{M}} \text{Scal}(x) |s(x)|^2 dv(x). \quad (*)$$

Here $\|s\|^2 = \int_{\tilde{M}} |s(x)|^2 dv(x)$ and $\|\nabla s\|^2 = \int_{\tilde{M}} |\nabla s(x)|^2 dv(x)$, where $|\nabla s(x)|^2 = \sum_{j=1}^m |\nabla_{e_j} s(x)|^2$. Indeed, $\langle D^2 s, s \rangle = \langle \nabla^* \nabla s, s \rangle + \frac{1}{4} \langle \text{Scal} s, s \rangle$, where $\langle \text{Scal} s, s \rangle = \int_{\tilde{M}} \text{Scal}(x) |s(x)|^2 dv(x)$. Since

$$\int_{\tilde{M}} \text{Scal}(x) |s(x)|^2 dv(x) \geq k \int_{\tilde{M}} |s(x)|^2 dv(x) = k \|s\|^2,$$

the identity $(*)$ gives

$$\langle D^2 s, s \rangle \geq 0 + \frac{k}{4} \|s\|^2 = \frac{k}{4} \|s\|^2.$$

Thus

$$D^2 \geq \frac{k}{4}, \quad \text{spec}(D) \cap \left(-\frac{\sqrt{k}}{2}, \frac{\sqrt{k}}{2} \right) = \emptyset.$$

Therefore D is invertible.

For the chiral spin Dirac operator D^+ on a closed even-dimensional spin manifold, write

$$S = S^+ \oplus S^-, \quad D = \begin{pmatrix} 0 & D^- \\ D^+ & 0 \end{pmatrix},$$

where $D^+ : C^\infty(M, S^+) \longrightarrow C^\infty(M, S^-)$. Then $\text{Index}(D^+) = \dim \ker D^+ - \dim \ker D^-$.

By the Atiyah–Singer index theorem,

$$\text{Index}(D^+) = \widehat{A}(M) = \left\langle \widehat{A}^{[m]}(TM), [M] \right\rangle, \quad [M] \in H_m(M).$$

PSC makes D invertible, so $\ker D^+ = \ker D^- = 0$. Thus $\widehat{A}(M) = \text{Index}(D^+) = 0$.

3.4 Quantitative higher index

Let $\widetilde{D}$ be the Dirac operator on the universal cover $\widetilde{M}$. Choose smooth functions $\varphi_{\varepsilon,0}$ and $\varphi_{\varepsilon,1}$ such that $\varphi_{\varepsilon,0}(\pm\infty) = \pm 1$ and $\varphi_{\varepsilon,1} = \varphi_{\varepsilon,0}^2 - 1$. We want $\varphi_{\varepsilon,0}(\lambda)^2 \approx 1$ and $\varphi_{\varepsilon,1}(\lambda) \approx 0$.

Choose a smooth nonzero bump function $\rho \in C_c^\infty(\mathbb{R})$ such that $\rho \geq 0$, ρ is even, $\int_{\mathbb{R}} \rho(\lambda) d\lambda = 1$, and $\text{supp}(\rho) \subseteq [-1/(6m), 1/(6m)]$. Let $\check{\rho}$ be the inverse Fourier transform. Then $\check{\rho}(0) = 1$ and $|\check{\rho}(\lambda)| < 1$ for $\lambda \neq 0$.

For a large even integer ε , define

$$C_\varepsilon = 2 \left(\int_{\mathbb{R}} \check{\rho}(\lambda)^\varepsilon d\lambda \right)^{-1}, \quad \varphi'_{\varepsilon,0}(\lambda) = C_\varepsilon \check{\rho}(\lambda)^\varepsilon.$$

Set $\varphi_{\varepsilon,0}(\lambda) = -1 + \int_{-\infty}^\lambda \varphi'_{\varepsilon,0}(t) dt$. Then $\int_{\mathbb{R}} \varphi'_{\varepsilon,0}(\lambda) d\lambda = 2$. Moreover,

$$\widehat{\varphi'_{\varepsilon,0}} = C_\varepsilon \underbrace{\rho * \cdots * \rho}_{\varepsilon \text{ factors}},$$

so

$$\text{supp}(\widehat{\varphi'_{\varepsilon,0}}) \subseteq \left[-\frac{\varepsilon}{6m}, \frac{\varepsilon}{6m} \right], \quad \text{supp}(\widehat{\varphi_{\varepsilon,1}}) \subseteq \left[-\frac{\varepsilon}{3m}, \frac{\varepsilon}{3m} \right].$$

Define the difference idempotent [MY26, (3.8)]

$$I_\varepsilon = \begin{pmatrix} \varphi_{\varepsilon,1}^+(\widetilde{D})^2 & -\varphi_{\varepsilon,1}^+(\widetilde{D})(1 - \varphi_{\varepsilon,1}^+(\widetilde{D}))\varphi_{\varepsilon,0}^-(\widetilde{D}) \\ -\varphi_{\varepsilon,1}^-(\widetilde{D})\varphi_{\varepsilon,0}^+(\widetilde{D}) & -\varphi_{\varepsilon,1}^-(\widetilde{D})^2 \end{pmatrix}.$$

Here $I_\varepsilon = P_\varepsilon - \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$, where P_ε is an idempotent.

Small propagation. We have $\text{prop}(I_\varepsilon) \leq \varepsilon/m$. Equivalently, $I_\varepsilon(x, y) = 0$ if $d(x, y) > \varepsilon/m$ [MY26, (3.9)].

Quantitative higher index [MY26, (3.13)].

$$[M] = \sum_{\sigma \in S_{m+1}} \text{sgn}(\sigma) \text{Tr}(I_\varepsilon(x_{\sigma(0)}, x_{\sigma(1)}) \cdots I_\varepsilon(x_{\sigma(m)}, x_{\sigma(0)})) [x_0, \dots, x_m].$$

The product is cyclic, the sum is antisymmetrized, and $m = \dim M$. The right-hand side is a cycle in the ε -homology group $H_m^\varepsilon(\widetilde{M}, \Gamma)$.

Definition 3.11 (ε -homology [MY26, (2.13)–(2.19)]). For $(x_0, \dots, x_k) \in \widetilde{M}^{k+1}$, define $\text{diam}(x_0, \dots, x_k) = \max_{i,j} d_{\widetilde{M}}(x_i, x_j)$. The ε -diagonal is

$$\text{diag}_\varepsilon(\widetilde{M}^{k+1}) = \{(x_0, \dots, x_k) : \text{diam}(x_0, \dots, x_k) \leq \varepsilon\}.$$

The chain group $C_k^\varepsilon(\widetilde{M}, \Gamma)$ consists of Γ -invariant measures supported in this set, where $\gamma(x_0, \dots, x_k) = (\gamma x_0, \dots, \gamma x_k)$. The boundary is

$$\partial \delta_{(x_0, \dots, x_k)} = \sum_{j=0}^k (-1)^j \delta_{(x_0, \dots, \widehat{x_j}, \dots, x_k)}, \quad \partial^2 = 0.$$

Thus the ε -homology group is $H_k^\varepsilon(\widetilde{M}, \Gamma) = H_k(C_*^\varepsilon(\widetilde{M}, \Gamma), \partial)$.

Intuition: it is generated by simplices of size at most ε .

3.5 The vanishing estimate

When ε is small, $H_k^\varepsilon(\widetilde{M}, \Gamma) \cong H_k(M)$. Choose an admissible rate $d > \psi_\Gamma$. Apply Schatten–Hölder and group decay to $[M]$. For sufficiently large i ,

$$\|M\| \leq C e^{d\varepsilon} \left\| (1 + \widetilde{D}^2)^i I_\varepsilon^* I_\varepsilon \right\|,$$

where the last norm is the operator norm on $L^2(\widetilde{M}, S_{\widetilde{M}})$, and $e^{d\varepsilon}$ comes from the exponential decay property of the fundamental group.

Assume $\text{Scal} \geq C > 0$. By Lichnerowicz,

$$\widetilde{D}^2 = \nabla^* \nabla + \frac{\text{Scal}}{4} \geq \frac{C}{4}.$$

Hence

$$\text{spec}(\widetilde{D}) \subseteq \left(-\infty, -\frac{\sqrt{C}}{2} \right] \cup \left[\frac{\sqrt{C}}{2}, \infty \right),$$

$$\text{Equivalently, } \text{spec}(\widetilde{D}) \cap \left(-\frac{\sqrt{C}}{2}, \frac{\sqrt{C}}{2} \right) = \emptyset.$$

Thus there is a spectral gap around 0.

We have $\varphi_{\varepsilon,1}(\widetilde{D}) = (\varphi_{\varepsilon,0}^2 - 1)(\widetilde{D})$. Choose $\varphi_{\varepsilon,0}$ so that $\varphi_{\varepsilon,0}^2$ is exponentially close to 1 for $|\lambda| \geq \sqrt{C}/2$.

Let $a = \sqrt{C}/2 > 0$ and $q = \sup_{|\lambda| \geq a} |\check{\rho}(\lambda)| < 1$. Then $|\check{\rho}(\lambda)|^\varepsilon \leq q^\varepsilon$ for $|\lambda| \geq a$, and

$$\sup_{|\lambda| \geq a} |\varphi_{\varepsilon,1}(\lambda)| \leq A \sqrt{\varepsilon} q^{\varepsilon-2}.$$

Let $q = e^{-b}$, where $b > 0$. Then

$$\sqrt{\varepsilon} q^{\varepsilon-2} = q^{-2} \sqrt{\varepsilon} e^{-b\varepsilon} \leq C e^{-b'\varepsilon}, \quad 0 < b' < b,$$

for sufficiently large ε . Thus

$$|\varphi_{\varepsilon,1}(\lambda)| \leq C e^{-c_m \varepsilon} \quad \text{for all } |\lambda| \geq \frac{\sqrt{C}}{2},$$

where c_m is a universal positive constant depending only on m .

The desired estimate is

$$\left\| (1 + \tilde{D}^2)^i I_\varepsilon^* I_\varepsilon \right\| \leq C' e^{-c_m \varepsilon}.$$

Indeed,

$$\begin{aligned} \left\| (1 + \tilde{D}^2)^i I_\varepsilon^* I_\varepsilon \right\| &\leq \sup_{|\lambda| \geq a} (1 + \lambda^2)^i \max \left\{ |\varphi_{\varepsilon,1}(\lambda)|^4, \right. \\ &\quad \left. |\varphi_{\varepsilon,1}(\lambda)(1 - \varphi_{\varepsilon,1}(\lambda))\varphi_{\varepsilon,0}(\lambda)|^2, |\varphi_{\varepsilon,1}(\lambda)\varphi_{\varepsilon,0}(\lambda)|^2 \right\} \\ &\leq C' \sup_{|\lambda| \geq a} (1 + \lambda^2)^i |\varphi_{\varepsilon,1}(\lambda)|^2 \\ &\leq C_i \varepsilon q^{2\varepsilon - 2N_i}. \end{aligned}$$

Here $N_i > 0$ depends only on i . Because

$$\begin{aligned} \|M\| &\leq C e^{d\varepsilon} \left\| (1 + \tilde{D}^2)^i I_\varepsilon^* I_\varepsilon \right\| \\ &\leq C e^{d\varepsilon} C' e^{-c_m \varepsilon} \\ &= C C' e^{(d - c_m)\varepsilon} \longrightarrow 0 \end{aligned}$$

as $\varepsilon \rightarrow \infty$, provided $d - c_m < 0$, we conclude that $\|M\| = 0$.

4 Higher index proof that the torus admits no positive scalar curvature

Let M be a complete Riemannian spin manifold, S the spinor bundle, and D the Dirac operator. The space $L^2(M; S)$ is the completion of $C_c^\infty(M; S)$ for the inner product $\langle u, v \rangle = \int_M \langle u(x), v(x) \rangle_x dx$, and

$$D: C_c^\infty(M; S) \subseteq L^2(M; S) \longrightarrow L^2(M; S).$$

Corollary 4.1 (Main result; [WY20, Corollary 3.3.5]). *The d -torus does not admit a metric of positive scalar curvature.*

For $u, v \in C_c^\infty(M; S)$, $\langle Du, v \rangle = \langle u, Dv \rangle$: D is formally self-adjoint. Also, $D^2 = \nabla^* \nabla + \kappa/4$, where ∇ is the spinor connection and κ is the scalar curvature [WY20, Theorem 3.1.5].

The graph is

$$\text{graph}(D) := \{(u, Du) : u \in C_c^\infty(M; S)\} \subseteq L^2(M; S) \oplus L^2(M; S).$$

Formal self-adjointness implies that D is closable; write $\overline{D}$ for its closure [WY20, Definitions D.1.2–D.1.3].

Definition 4.2 ([WY20, Definition D.1.4; Definition 8.1.6]). The minimal domain is $\mathcal{D}_{\min} = \text{Dom}(\overline{D})$. The maximal domain $\mathcal{D}_{\max}$ is the collection of all $v \in L^2(M; S)$ such that there exists $w \in L^2(M; S)$ such that, for all $u \in C_c^\infty(M; S)$, $\langle Du, v \rangle = \langle u, w \rangle$. The operator D is essentially self-adjoint if $\mathcal{D}_{\min} = \mathcal{D}_{\max}$.

A formally self-adjoint operator D is essentially self-adjoint if, whenever $v, w \in L^2(M; S)$ satisfy $\langle Du, v \rangle = \langle u, w \rangle$ for all $u \in C_c^\infty(M; S)$, there exists a sequence (v_n) in $C_c^\infty(M; S)$ such that $v_n \rightarrow v$ and $Dv_n \rightarrow w$ in L^2 -norm.

In local coordinates, if $D = \sum_j a_j(x) \frac{\partial}{\partial x_j} + b(x)$, then its symbol is $\sigma_D(x, \xi) = \sum_j \xi_j a_j(x)$ [WY20, Definition 8.1.3]. For the spin Dirac operator,

$$\sigma_D(x, \xi)^* = -\sigma_D(x, \xi), \quad \sigma_D(x, \xi)^2 = -\|\xi\|^2 1,$$

so $\sigma_D(x, \xi)^* \sigma_D(x, \xi) = \|\xi\|^2 1$. Thus, by [WY20, Definition 8.1.7],

$$c_D = \sup_{\substack{x \in M, \xi \in T_x^* M \\ \|\xi\|=1}} \|\sigma_D(x, \xi)\| = 1.$$

Proposition 4.3 ([WY20, Proposition 8.1.8]). *Let D be a formally self-adjoint first order differential operator. Assume that either the support of D is compact, or M is complete and c_D is finite. Then D is essentially self-adjoint.*

Consequently, for the spin Dirac operator:

Proposition 4.4 ([WY20, Proposition 3.2.1]). *Assume that the Riemannian metric on M is complete. Then D is essentially self-adjoint.*

In what follows, D denotes its self-adjoint closure when applying functional calculus or considering its spectrum and resolvent [WY20, Remark D.1.8]. Then, for any continuous bounded function $\phi: \mathbb{R} \rightarrow \mathbb{C}$, the functional calculus lets us define a bounded normal operator $\phi(D)$ on $L^2(M; S)$ [WY20, Theorem D.1.7].

Proposition 4.5 ([WY20, Proposition 3.2.4]).

- (1) $\text{supp } \widehat{\phi} \subseteq [-r, r] \implies \text{prop}(\phi(D)) \leq r$, where $\widehat{\phi}$ is the Fourier transform of ϕ .
- (2) If ϕ is compactly supported, then $\phi(D)$ is locally compact.

Definition 4.6 ([WY20, Definition 3.2.6]). The Roe algebra $C^*(M)$ of $L^2(M; S)$ is the C^* -algebra generated by all locally compact, finite propagation operators on $L^2(M; S)$.

If $\phi \in C_0(\mathbb{R})$, then $\phi(D) \in C^*(M)$. Let $\psi: \mathbb{R} \rightarrow [-1, 1]$ be continuous and odd, with $\lim_{x \rightarrow \pm\infty} \psi(x) = \pm 1$. Set $F = \psi(D)$. Then $\text{Ind}[F] \in K_i(C^*(M))$, where $i \equiv \dim M \pmod{2}$ [WY20, Lemma 3.2.7].

Theorem 4.7 ([WY20, Theorem 3.2.8]). If M is a complete spin Riemannian manifold with scalar curvature bounded below by a positive constant, then $\text{Ind}[F] = 0$.

Proof. For $u \in C_c^\infty(M; S)$,

$$\begin{aligned} \|Du\|^2 &= \langle D^2u, u \rangle \\ &= \|\nabla u\|^2 + \frac{1}{4} \int_M \kappa(x) |u(x)|^2 dx \\ &\geq \frac{1}{4} (\inf_M \kappa) \|u\|^2. \end{aligned}$$

Thus $\|Du\| \geq \frac{1}{2} \sqrt{\inf_M \kappa} \|u\|$ for $u \in \text{Dom}(\overline{D})$. For real λ with $|\lambda| < \frac{1}{2} \sqrt{\inf_M \kappa}$,

$$\begin{aligned} \|(D - \lambda)u\| &\geq \|Du\| - |\lambda| \|u\| \\ &\geq \left(\frac{1}{2} \sqrt{\inf_M \kappa} - |\lambda| \right) \|u\|. \end{aligned}$$

Hence $D - \lambda$ is injective and has closed range. Since D is self-adjoint and λ is real, $\text{Range}(D - \lambda)^\perp = \ker(D - \lambda) = \{0\}$. Therefore $\text{Range}(D - \lambda) = L^2(M; S)$, and

$$\|(D - \lambda)^{-1}\| \leq \left(\frac{1}{2} \sqrt{\inf_M \kappa} - |\lambda| \right)^{-1}.$$

Choose $0 < c < \frac{1}{2} \sqrt{\inf_M \kappa}$. Then $\text{spec}(D) \cap [-c, c] = \emptyset$. Choose ψ as above with $\psi^2 = 1$ on the spectrum of D , without changing the index [WY20, Lemma 3.2.7]. Set $F = \psi(D)$. Then $F^2 = (\psi^2)(D) = 1$, so $\text{Ind}[F] = 0$ in $K_i(C^*(M))$ [WY20, Lemma 2.8.7]. $\square$

Proof of Corollary 4.1. Apply the construction to the universal cover of the torus. Let $M = (S^1)^d$ carry an arbitrary Riemannian metric and spin structure. Let $\tilde{M}$ be the universal cover with the complete lifted metric, $\tilde{S}$ its spinor bundle, and $\tilde{D}$ its Dirac operator. Let $\chi: \mathbb{R} \rightarrow [-1, 1]$ be continuous and odd, with $\lim_{x \rightarrow \pm\infty} \chi(x) = \pm 1$. Set $\tilde{F} = \chi(\tilde{D})$; then $\text{Ind}[\tilde{F}] \in K_i(C^*(\tilde{M}))$. For $t \in [1, \infty)$, let $\tilde{F}_t := \chi(t^{-1}\tilde{D})$. The family $(\tilde{F}_t)$ is a multiplier of the localisation algebra $C_L^*(\tilde{M})$ [WY20, Definition 3.3.1], such that $(1 - \tilde{F}_t^2) \in C_L^*(\tilde{M})$. Thus $\text{Index}_L(\tilde{F}) \in K_i(C_L^*(\tilde{M}))$ [WY20, Lemma 3.3.2].

The group $K_i(C_L^*(\tilde{M})) \cong \mathbb{Z}$, and $\text{Index}_L(\tilde{F})$ is a generator [WY20, Proposition 3.3.3]. Hence $\text{Index}_L(\tilde{F}) \neq 0$. Evaluation at one is the $*$ -homomorphism

$$\begin{aligned} \text{ev}: C_L^*(\tilde{M}) &\longrightarrow C^*(\tilde{M}), \\ (T_t) &\longmapsto T_1. \end{aligned}$$

It satisfies

$$\text{ev}_*(\text{Index}_L(\tilde{F})) = \text{Ind}[\tilde{F}].$$

Since ev_* is injective [WY20, Theorem 3.3.4], $\text{Ind}[\tilde{F}] \neq 0$. If M had positive scalar curvature, its compactness would give $\inf_M \kappa > 0$, and the complete lifted metric would have the same lower bound. Theorem 4.7 would give $\text{Ind}[\tilde{F}] = 0$, a contradiction. So the d -torus admits no metric of positive scalar curvature. $\square$

5 Sharp estimates and the Dirac operator

Theorem 5.1 ([Lla98, Theorem 4.1]). *Let M be a compact connected Riemannian spin manifold without boundary of dimension $n \geq 2$, with normalized scalar curvature $\tilde{\kappa}_g = \kappa_g / (n(n-1))$. Suppose there exists a 1-contracting map $f: (M, g) \rightarrow (S^n, g_0)$ of nonzero degree. Then there exists $x \in M$ with $\tilde{\kappa}_g(x) < 1$, or $M \equiv S^n$ and f is an isometry.*

Proof. First assume that $\tilde{\kappa}_g(x) \geq 1$ for all $x \in M$. We need to show that f is an isometry.

Even dimensions: $\dim M = 2n$

S^{2n} is the unit $2n$ -sphere with g_0 , and $f: M \rightarrow S^{2n}$. Assume $\tilde{\kappa}_g > 1$, equivalently $\kappa > 2n(2n-1)$. For the spinor bundles S and E_0 of [Lla98, Section 2], let $E = f^*E_0$ and $E^+ = f^*E_0^+$; consider $S \otimes E^+$. The operator D_E preserves $\Gamma(S \otimes E^+)$, and $D_{E^+} = D_E|_{\Gamma(S \otimes E^+)}$.

For $p \in M$, $f_*: T_p M \rightarrow T_{f(p)} S^{2n}$ and $(f_*)^* f_*: T_p M \rightarrow T_p M$, where

$$g((f_*)^* f_* v, w) = g_0(f_* v, f_* w).$$

Choose a g -orthonormal tangent frame $\{e_1, \dots, e_{2n}\}$ and a g_0 -orthonormal frame $\{\varepsilon_1, \dots, \varepsilon_{2n}\}$ such that $f_* e_j = \|f_* e_j\|_{g_0} \varepsilon_j$ at p . Then, for $i \neq j$,

$$g_0(f_* e_i, f_* e_j) = g(e_i, (f_*)^* f_* e_j) = 0.$$

When $f_* e_j \neq 0$, let $\varepsilon_j = f_* e_j / \|f_* e_j\|_{g_0}$ and $\lambda_j = 1 / \|f_* e_j\|_{g_0}$. Then $\varepsilon_j = \lambda_j f_* e_j$, and

$$\delta_{ij} = g_0(\varepsilon_i, \varepsilon_j) = \lambda_i \lambda_j g_0(f_* e_i, f_* e_j) \quad (f_* e_i, f_* e_j \neq 0).$$

So

$$1 = g_0(\varepsilon_j, \varepsilon_j) = g_0(\lambda_j f_* e_j, \lambda_j f_* e_j) = \lambda_j^2 g_0(f_* e_j, f_* e_j) \leq \lambda_j^2.$$

Also, $(\nabla e_j)_p = 0$ and $(\nabla \varepsilon_j)_{f(p)} = 0$. In the formulas below, $1/\lambda_j$ denotes $\|f_* e_j\|_{g_0}$, including when $f_* e_j = 0$.

Bochner formula. For all $\phi, \psi \in \Gamma(S \otimes E)$, write $\langle \phi, \psi \rangle = \int_M g_x(\phi, \psi) d \text{vol}_g$. Then

$$D_E^2 \phi = \nabla^* \nabla \phi + \frac{1}{4} \kappa \phi + \mathcal{R}^E \phi, \quad \langle D_E^2 \phi, \phi \rangle = \|D_E \phi\|^2, \quad \langle \nabla^* \nabla \phi, \phi \rangle = \|\nabla \phi\|^2.$$

Thus

$$\|D_E \phi\|^2 = \|\nabla \phi\|^2 + \frac{1}{4} \int_M \kappa(x) g_x(\phi, \phi) d \text{vol}_g + \langle \mathcal{R}^E \phi, \phi \rangle.$$

Furthermore,

$$\mathcal{R}^E(\sigma \otimes v) = \frac{1}{2} \sum_{i,j=1}^{2n} (e_i \cdot e_j \cdot \sigma) \otimes (R_{e_i e_j}^E v), \quad R_{e_i e_j}^E = R_{f_* e_i f_* e_j}^{E_0}.$$

$$R_{e_i e_j}^E = \frac{1}{4} \sum_{k,\ell=1}^{2n} [g_0(f_* e_i, \varepsilon_\ell) g_0(f_* e_j, \varepsilon_k) - g_0(f_* e_j, \varepsilon_\ell) g_0(f_* e_i, \varepsilon_k)] \varepsilon_k \cdot \varepsilon_\ell.$$

We have $f_*e_i = \varepsilon_i/\lambda_i$ and $f_*e_j = \varepsilon_j/\lambda_j$. By orthonormality,

$$g_0(f_*e_i, \varepsilon_\ell) = \frac{\delta_{i\ell}}{\lambda_i}, \quad g_0(f_*e_j, \varepsilon_k) = \frac{\delta_{jk}}{\lambda_j}.$$

Hence, for $i \neq j$,

$$\begin{aligned} R_{e_i e_j}^E &= \frac{1}{4\lambda_i \lambda_j} \sum_{k, \ell=1}^{2n} (\delta_{i\ell} \delta_{jk} - \delta_{j\ell} \delta_{ik}) \varepsilon_k \cdot \varepsilon_\ell \\ &= \frac{1}{4\lambda_i \lambda_j} (\varepsilon_j \cdot \varepsilon_i - \varepsilon_i \cdot \varepsilon_j) \\ &= \frac{1}{2\lambda_i \lambda_j} \varepsilon_j \cdot \varepsilon_i. \end{aligned}$$

So,

$$\mathcal{R}^E(\sigma \otimes v) = \frac{1}{4} \sum_{i \neq j} \frac{1}{\lambda_i \lambda_j} (e_i \cdot e_j \cdot \sigma) \otimes (\varepsilon_j \cdot \varepsilon_i \cdot v).$$

Choose a basis $\{\sigma_\alpha\}_{\alpha=1}^{2^{2n}}$ for S and a basis $\{v_\beta\}_{\beta=1}^{2^{2n}}$ for E_0 . For $\phi \in \Gamma(S \otimes E)$, write $\phi = \sum_{\alpha, \beta} a_{\alpha\beta} \sigma_\alpha \otimes v_\beta$. Then

$$\begin{aligned} \mathcal{R}^E \phi &= \frac{1}{4} \sum_{i \neq j} \sum_{\alpha, \beta} \frac{a_{\alpha\beta}}{\lambda_i \lambda_j} (e_i \cdot e_j \cdot \sigma_\alpha) \otimes (\varepsilon_j \cdot \varepsilon_i \cdot v_\beta), \\ g_p(\mathcal{R}^E \phi, \phi) &= \frac{1}{4} \sum_{i \neq j} \sum_{k, \ell} \sum_{\alpha, \beta} \frac{a_{\alpha\beta} \overline{a_{k\ell}}}{\lambda_i \lambda_j} \\ &\quad \cdot g_p(e_i \cdot e_j \cdot \sigma_\alpha, \sigma_k) g_p(\varepsilon_j \cdot \varepsilon_i \cdot v_\beta, v_\ell). \end{aligned}$$

For fixed $i \neq j$, consider $e_i \cdot e_j$ on S_p and $\varepsilon_j \cdot \varepsilon_i$ on E_p . We have

$$\begin{aligned} (e_i \cdot e_j)^2 &= e_i \cdot e_j \cdot e_i \cdot e_j = -e_i \cdot e_i \cdot e_j \cdot e_j = -1, \\ (e_i \cdot e_j)^* &= (-e_j) \cdot (-e_i) = -e_i \cdot e_j, \quad (e_i \cdot e_j)^*(e_i \cdot e_j) = 1. \end{aligned}$$

Also, $\sigma \mapsto e_i \cdot e_j \cdot \sigma \mapsto -\sigma$, and

$$g_p(\sigma, e_i \cdot e_j \cdot \sigma) = -\overline{g_p(\sigma, e_i \cdot e_j \cdot \sigma)}.$$

Construct an orthonormal complex basis $\{\sigma_\alpha, e_i \cdot e_j \cdot \sigma_\alpha\}_{\alpha=1}^{2^{2n-1}}$ for S , and $\{v_\beta, \varepsilon_j \cdot \varepsilon_i \cdot v_\beta\}_{\beta=1}^{2^{2n-1}}$ for E_p . For $\phi \in \Gamma(S \otimes E)$,

$$\begin{aligned} \phi &= \sum_{\alpha, \beta} (a_{\alpha\beta} \sigma_\alpha \otimes v_\beta + b_{\alpha\beta} \sigma_\alpha \otimes \varepsilon_j \cdot \varepsilon_i \cdot v_\beta \\ &\quad + c_{\alpha\beta} e_i \cdot e_j \cdot \sigma_\alpha \otimes v_\beta + d_{\alpha\beta} e_i \cdot e_j \cdot \sigma_\alpha \otimes \varepsilon_j \cdot \varepsilon_i \cdot v_\beta). \end{aligned}$$

Lemma 5.2 ([Lla98, Lemma 4.5]). *For $i \neq j$, $g_p(((e_i \cdot e_j) \otimes (\varepsilon_j \cdot \varepsilon_i))\phi, \phi) \geq -g_p(\phi, \phi)$.*

Then

$$\begin{aligned} g_P(\mathcal{R}^E \phi, \phi) &\geq -\frac{1}{4} \sum_{i \neq j} \underbrace{\frac{1}{\lambda_i \lambda_j}}_{\leq 2n(2n-1)} g_P(\phi, \phi) \\ &\geq -\frac{1}{4} 2n(2n-1) g_P(\phi, \phi). \end{aligned}$$

Consequently,

$$\begin{aligned} \|D_E \phi\|^2 &\geq \|\nabla \phi\|^2 + \frac{1}{4} \int_M \left(\kappa - \sum_{i \neq j} \frac{1}{\lambda_i \lambda_j} \right) g_x(\phi, \phi) d \text{vol}_g \\ &\geq \|\nabla \phi\|^2 + \frac{1}{4} \int_M (\kappa - 2n(2n-1)) g_x(\phi, \phi) d \text{vol}_g \\ &= \|\nabla \phi\|^2 + \frac{2n(2n-1)}{4} \int_M (\tilde{\kappa}_g - 1) g_x(\phi, \phi) d \text{vol}_g. \end{aligned}$$

By assumption, $\tilde{\kappa}_g > 1$. Compactness gives $\min_M (\tilde{\kappa}_g - 1) > 0$.

$$\implies \|D_E \phi\|^2 \geq \frac{2n(2n-1)}{4} \min_M (\tilde{\kappa}_g - 1) \|\phi\|^2.$$

If $D_E \phi = 0$, then $\phi = 0$, so $\ker D_E = \{0\} = \ker D_E^2$.

$$D_E^2 \phi = 0 \implies 0 = \langle D_E^2 \phi, \phi \rangle = \|D_E \phi\|^2 \implies D_E \phi = 0.$$

Analytic: $\ker D_{E^+} = \{0\}$.

$$\begin{aligned} D_{E^+}^\pm : \Gamma(S^\pm \otimes E^+) &\longrightarrow \Gamma(S^\mp \otimes E^+). \\ D_{E^+} \phi &= \underbrace{D_{E^+}^+ \phi^+ + D_{E^+}^- \phi^-}_{(S^- \otimes E^+) \oplus (S^+ \otimes E^+)} \implies \ker D_{E^+} = \ker D_{E^+}^+ \oplus \ker D_{E^+}^-. \\ \text{Index}(D_{E^+}) &= \dim \ker(D_{E^+}^+) - \dim \ker(D_{E^+}^-) = 0. \end{aligned}$$

Topology:

$$\text{Index}(D_{E^+}) = 2^n \{ \text{ch}(E^+) \hat{A}(M) \} [M].$$

$H^{2j}(S^{2n}; \mathbb{Q}) = 0$ for $0 < j < n$.

$$\begin{aligned} \text{ch}(E_0^+) &= 2^{2n-1} + \text{ch}_n(E_0^+), \\ \implies \text{ch}(E^+) &= 2^{2n-1} + f^* \text{ch}_n(E_0^+). \end{aligned}$$

$$\text{ch}_n(E_0^+) = \frac{(-1)^{n-1}}{(n-1)!} c_n(E_0^+).$$

$$\{ \text{ch}(E^+) \hat{A}(M) \} [M] = 2^{2n-1} \{ \hat{A}(M) \} [M] + \{ f^* \text{ch}_n(E_0^+) \} [M].$$

Since $\kappa > 0$, $\{\widehat{A}(M)\}[M] = 0$. Thus

$$\begin{aligned} \text{Index}(D_{E^+}) &= 2^n \{f^* \text{ch}_n(E_0^+)\}[M] \\ &= 2^n \deg(f) \{\text{ch}_n(E_0^+)\}[S^{2n}]. \end{aligned}$$

$$\begin{aligned} \text{ch}_n(E_0^+) &= \frac{1}{n!} \left(\frac{i}{2\pi} \right)^n \text{tr}_{E_0^+}((R^{E_0})^n) \\ &= \frac{1}{n!} \left(\frac{i}{2\pi} \right)^n \frac{(-1)^n (2n)!}{4^n} i^{-n} 2^{2n-1} d \text{vol}_{g_0} \\ &= \frac{(-1)^n (2n)!}{2 n! (2\pi)^n} d \text{vol}_{g_0} \neq 0. \end{aligned}$$

$$\text{Index}(D_{E^+}) \neq 0.$$

Now assume $\widetilde{\kappa}_g \geq 1$ everywhere. $\kappa \geq 2n(2n-1) > 0$, so the topological index still applies. Let $0 \neq \phi \in \Gamma(S \otimes E^+)$, $D_{E^+} \phi = D_E \phi = 0$.

$$\begin{aligned} 0 = \|D_E \phi\|^2 &\geq \|\nabla \phi\|^2 + \frac{1}{4} \int_M \left(\kappa - \sum_{i \neq j} \frac{1}{\lambda_i \lambda_j} \right) g_x(\phi, \phi) d \text{vol}_g \\ &\geq \|\nabla \phi\|^2 + \frac{1}{4} \int_M \left[\kappa - 2n(2n-1) + \sum_{i \neq j} \left(1 - \frac{1}{\lambda_i \lambda_j} \right) \right] g_x(\phi, \phi) d \text{vol}_g \geq 0. \end{aligned}$$

Hence $\nabla \phi = 0$ everywhere. Then

$$Xg(\phi, \phi) = g(\nabla_X \phi, \phi) + g(\phi, \nabla_X \phi) = 0, \quad X \text{ a vector field.}$$

$g(\phi, \phi)$ is a positive constant on M .

$$0 \leq 1 - \frac{1}{\lambda_i \lambda_j} \leq 0 \quad \forall i \neq j \implies \lambda_i = 1 \quad \forall i.$$

$g_0(f_* e_i, f_* e_j) = \delta_{ij} = g(e_i, e_j)$, so f is a local isometry. M is connected, and S^{2n} is simply connected for $2n \geq 2$. The covering has one sheet, so f is an isometry.

Odd dimensional: $\dim M = 2n - 1 \geq 3$

$f: (M, g) \longrightarrow (S^{2n-1}, g_0)$ is a 1-contracting map of nonzero degree. Consider

$$\begin{array}{ccccc} M \times S_r^1 & \xrightarrow{f \times \frac{1}{r} \text{id}} & S^{2n-1} \times S^1 & \xrightarrow{h} & S^{2n-1} \wedge S^1 \cong S^{2n} \\ \downarrow & & \downarrow & & \downarrow \\ g + ds^2 & & g_0 + ds^2 & & \widetilde{g}. \end{array}$$

Here h is a 1-contracting map of nonzero degree, and $\tilde{g}$ is the standard metric on the unit sphere S^{2n} , as in [Lla98, Section 4]. $\tilde{f} = h \circ (f \times \frac{1}{r} \text{id})$. $\frac{1}{r} \text{id}: S_r^1 \rightarrow S^1$ has degree 1. $\deg(\tilde{f}) = \deg(h) \deg(f) \neq 0$.

$$\begin{aligned} \|\tilde{f}_*(v, t)\|^2 &= \left\| h_* \left(f_* v, \frac{t}{r} \right) \right\|^2 \\ &\leq \|f_* v\|^2 + \frac{1}{r^2} \|t\|^2 \\ &\leq \|v\|^2 + \|t\|^2 \quad \text{for } r > 1. \end{aligned}$$

Thus $\tilde{f}$ is 1-contracting. Construct S over $M \times S_r^1$, E_0 over S^{2n} . $E = \tilde{f}^* E_0^+$. Choose an orthonormal frame

$$\underbrace{e_1, \dots, e_{2n-1}}_{\text{tangent to } M} \quad \underbrace{e_{2n}}_{\text{tangent to } S_r^1} \quad .$$

$$R_{e_i e_j}^E = \frac{1}{4} (\tilde{f}_* e_j \cdot \tilde{f}_* e_i - \tilde{f}_* e_i \cdot \tilde{f}_* e_j), \quad \|R_{e_i e_j}^E\| \leq \frac{1}{2} \|\tilde{f}_* e_i\| \|\tilde{f}_* e_j\|.$$

By contraction, for $1 \leq i \leq 2n-1$,

$$\|\tilde{f}_* e_i\| \leq \|f_* e_i\| \leq 1, \quad \|\tilde{f}_* e_{2n}\| \leq \frac{1}{r}.$$

$$\begin{aligned} g_x(\mathcal{R}^E \phi, \phi) &\geq -\frac{1}{4} \sum_{1 \leq i \neq j \leq 2n-1} \|f_* e_i\| \|f_* e_j\| g_x(\phi, \phi) - \frac{1}{2r} \sum_{i=1}^{2n-1} \|f_* e_i\| g_x(\phi, \phi) \\ &\geq -\left[\frac{1}{4} (2n-1)(2n-2) + \frac{2n-1}{2r} \right] g_x(\phi, \phi). \end{aligned}$$

$$\langle \mathcal{R}^E \phi, \phi \rangle \geq -\frac{1}{4} (2n-1)(2n-2) \|\phi\|^2 - \frac{2n-1}{2r} \|\phi\|^2.$$

Recall: $D_E^2 = \nabla^* \nabla + \frac{1}{4} \kappa + \mathcal{R}^E$. Then

$$\begin{aligned} \langle D_E^2 \phi, \phi \rangle &= \|D_E \phi\|^2 \\ &= \|\nabla \phi\|^2 + \frac{1}{4} \int_{M \times S_r^1} \kappa g_x(\phi, \phi) d \text{vol} + \langle \mathcal{R}^E \phi, \phi \rangle, \end{aligned}$$

where $\kappa = (2n-1)(2n-2)\tilde{\kappa}$. Thus

$$\|D_E \phi\|^2 \geq \|\nabla \phi\|^2 + \frac{1}{4} (2n-1)(2n-2) \int_{M \times S_r^1} \left[\tilde{\kappa} - 1 - \frac{2}{(2n-2)r} \right] g_x(\phi, \phi) d \text{vol}.$$

If $\tilde{\kappa} > 1$, then $\min_M (\tilde{\kappa} - 1) > 0$. Choose

$$r > \max \left\{ 1, \frac{2}{(2n-2) \min_M (\tilde{\kappa} - 1)} \right\}.$$

Then $\ker D_E = \{0\}$, so $\ker D_E^+ = \ker D_E^- = \{0\}$, and $\text{Index}(D_E) = 0$ (Analytical). But from Atiyah–Singer, $\text{Index}(D_E) \neq 0$ ($\tilde{\kappa} > 1$ everywhere).

For $\tilde{\kappa} \geq 1$: Wang–Xie (2023) [WX23]. (Topological)

$$\text{Index}(D_E) = 2^n \{\text{ch}(E) \widehat{A}(M \times S_r^1)\} [M \times S_r^1], \quad \{\widehat{A}(M \times S_r^1)\} [M \times S_r^1] = 0 \quad \text{since } \kappa > 0.$$

On the sphere,

$$\begin{aligned} \text{Index}(D_E) &= 2^n \{\tilde{f}^* \text{ch}_n(E_0^+)\} [M \times S_r^1] \\ &= 2^n \deg(\tilde{f}) \{\text{ch}_n(E_0^+)\} [S^{2n}] \neq 0, \quad \tilde{\kappa} \geq 1. \end{aligned}$$

So there exists $0 \neq \phi \in \Gamma(S \otimes E)$ with $D_E \phi = 0$. The map f is 1-contracting, hence $(1, \Lambda^2)$ -contracting, and $\kappa = (2n-1)(2n-2)\tilde{\kappa} \geq (2n-1)(2n-2)$. By Wang–Xie [WX23, Theorem 3.2], $f: (M, g) \rightarrow (S^{2n-1}, g_0)$ is an isometry. $\square$

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